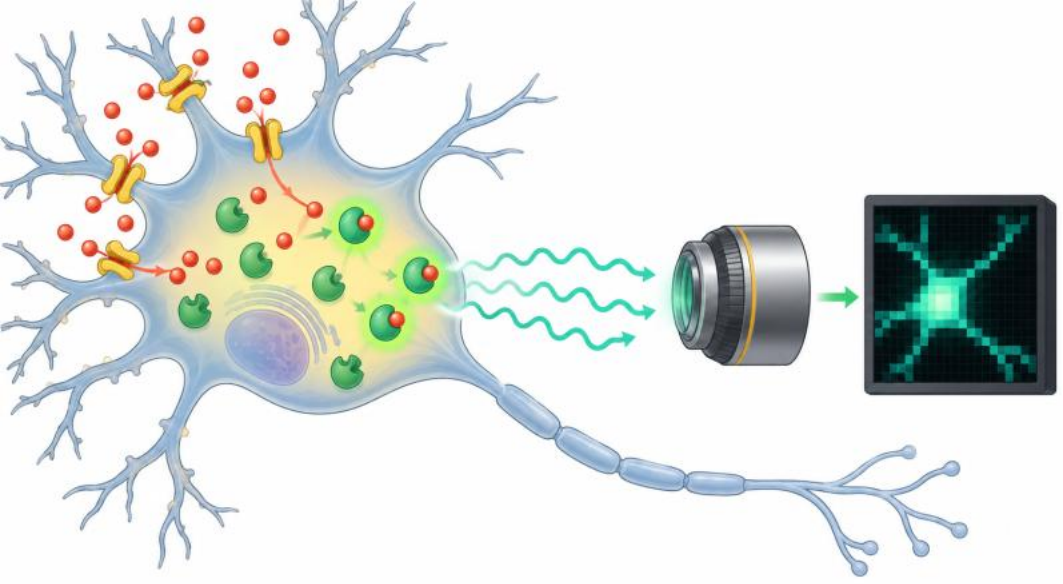
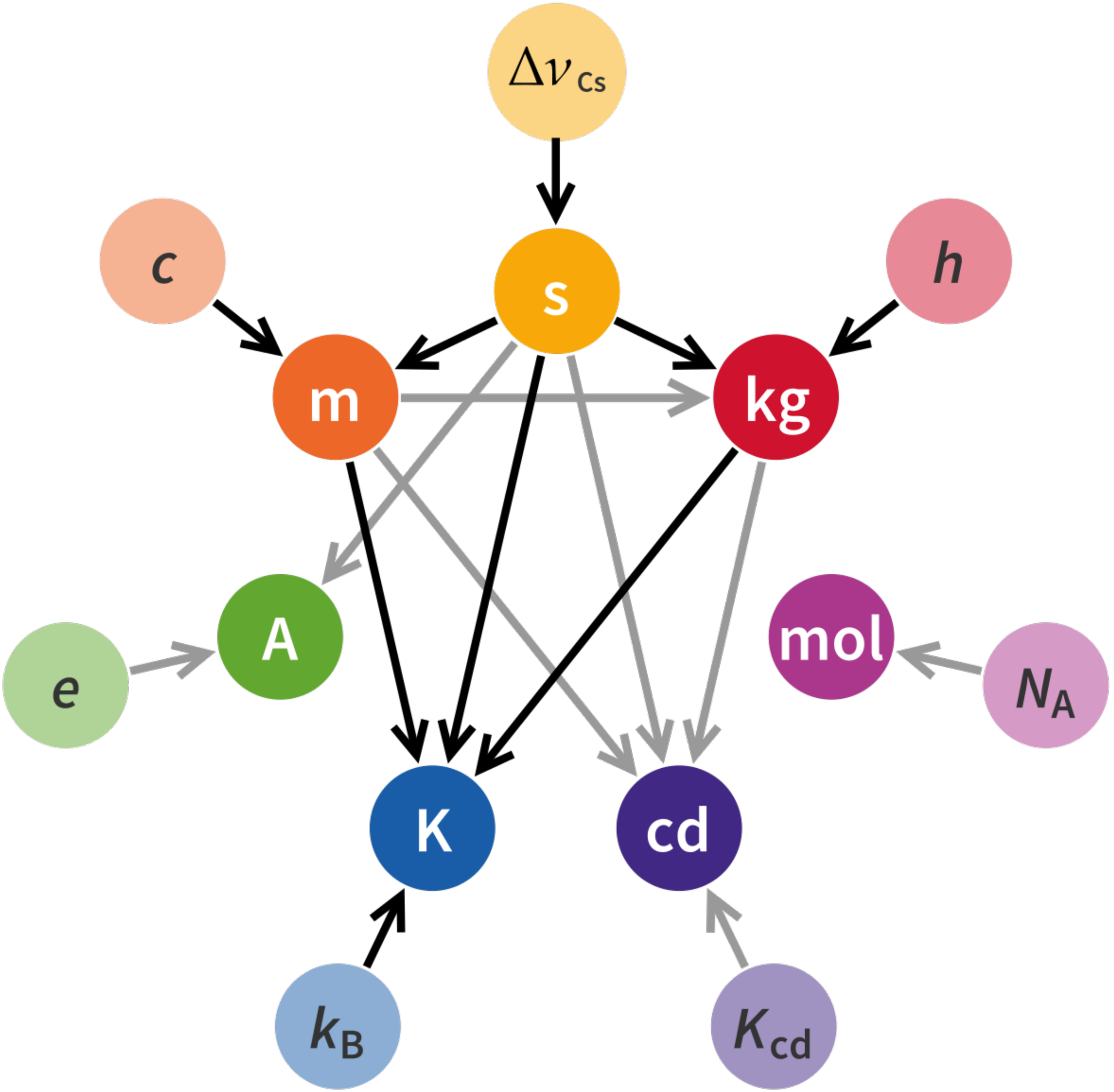
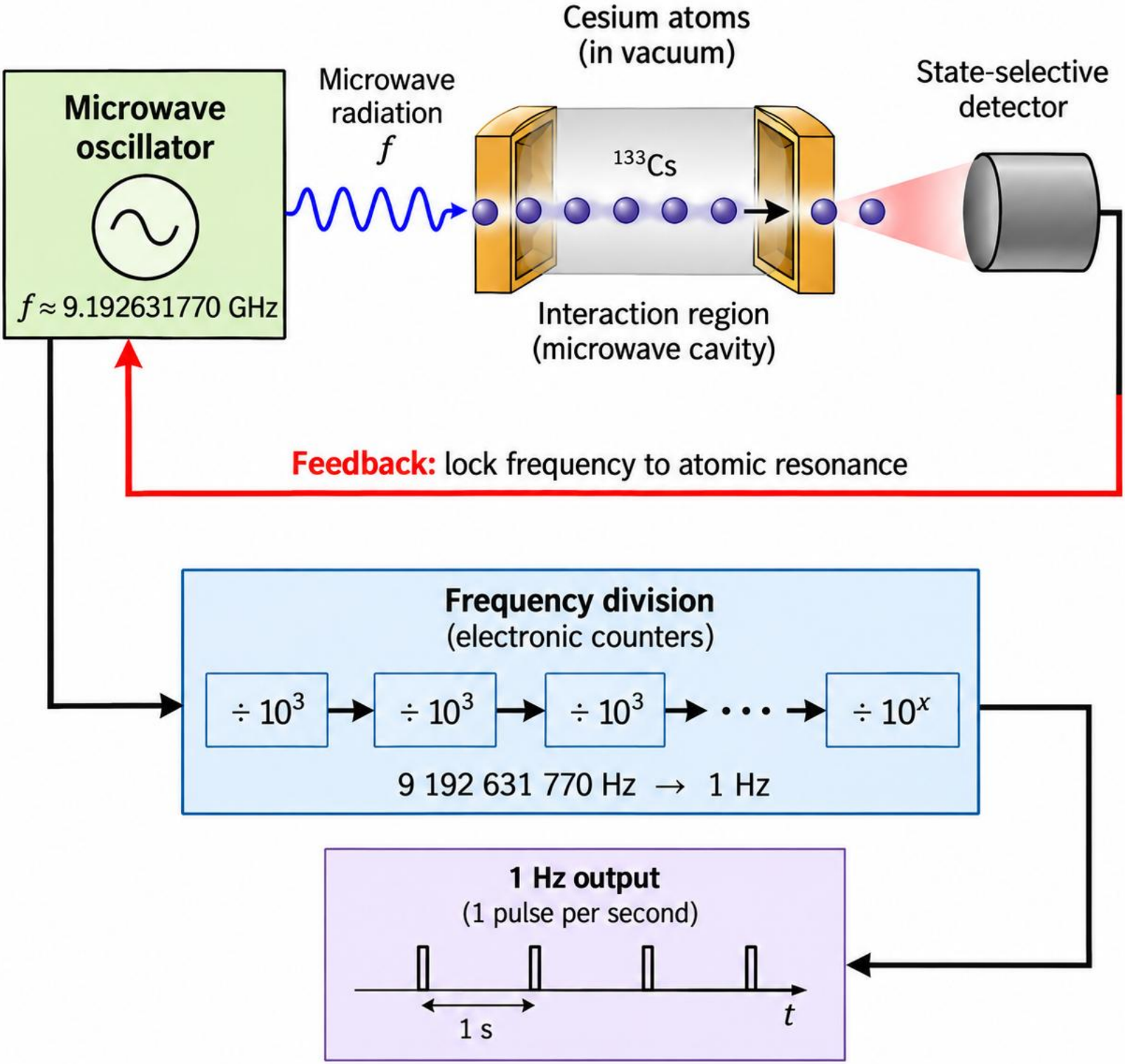
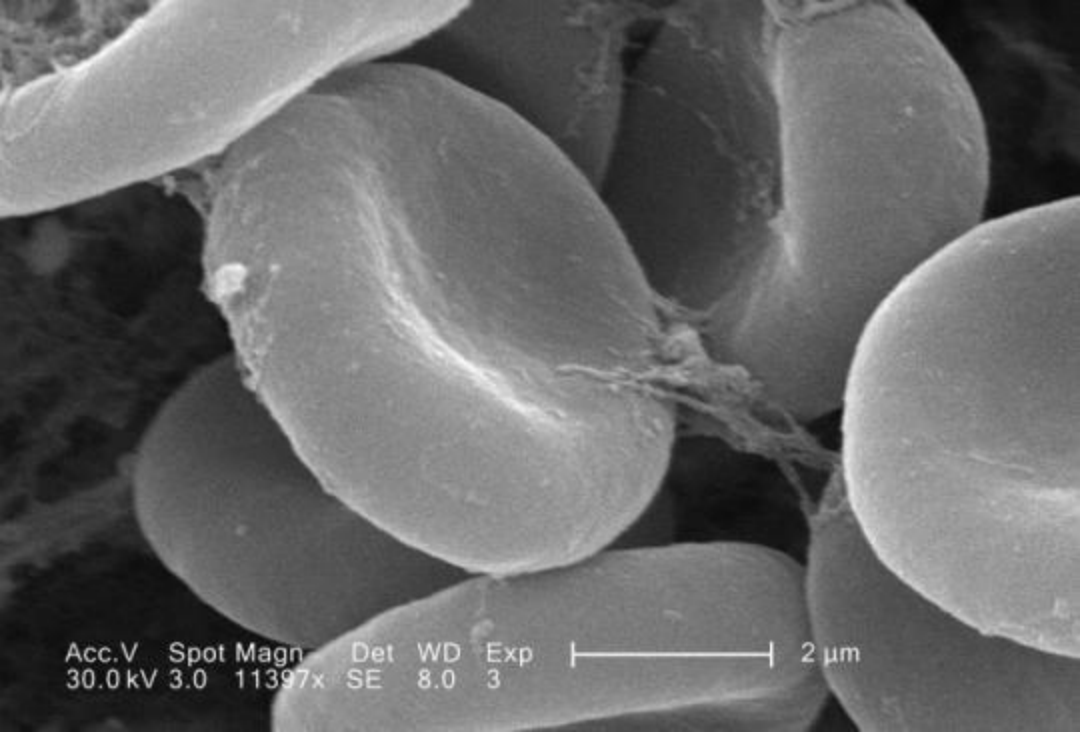
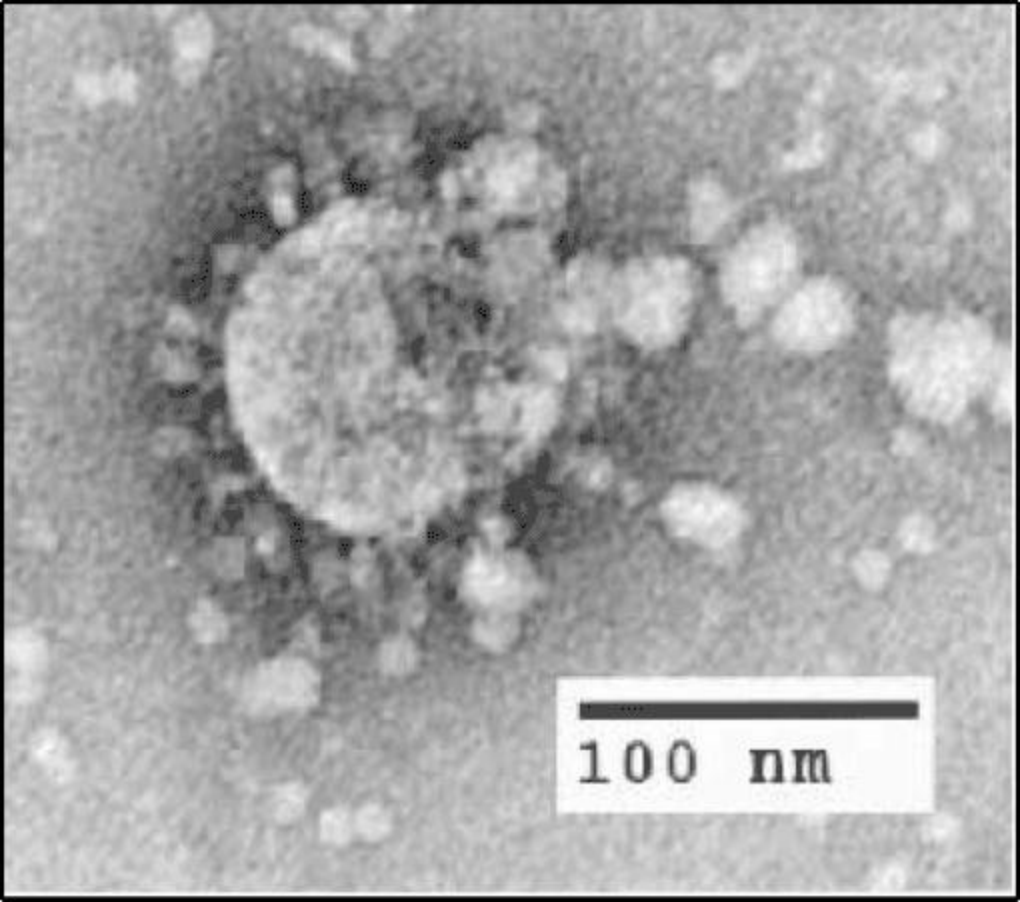

TARGET QUANTITY
Quantity intended to be measured
- Question: “What do I want to know?”
MEASUREMENT
Experimental process that produces information about the measurand
- Question: “How do I determine it?”
RESULT
Value attributed to the measurand
Reported with unit and uncertainty

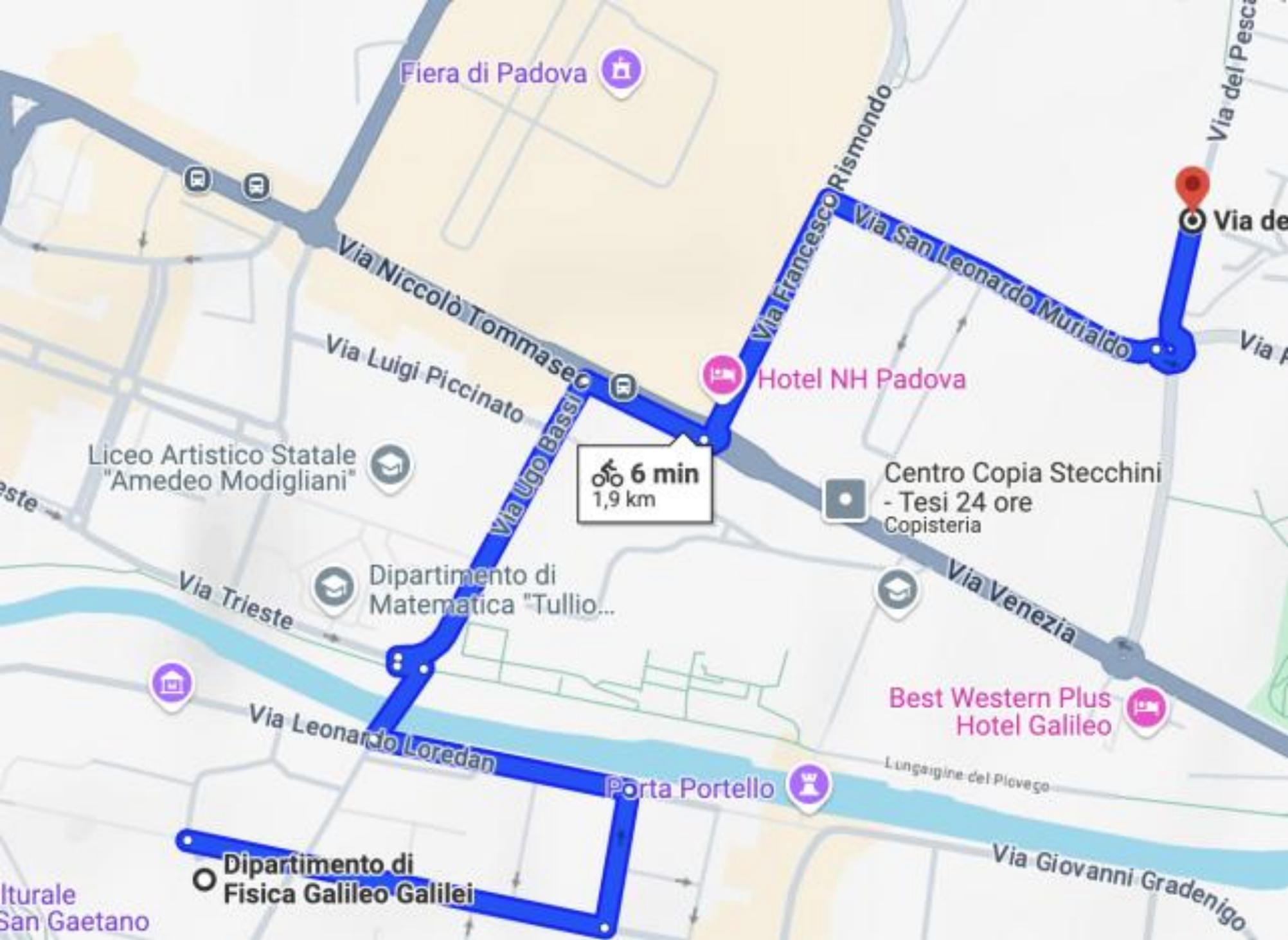



$\Delta\nu_{Cs}$
c
s
h
m
kg
e
A
mol
N_A
K
cd
k_B
K_{cd}


Cesium atoms
(in vacuum)
Microwave oscillator
Microwave radiation
f
State-selective detector
^{133}Cs
$f \approx 9.192631770$ GHz
Interaction region
(microwave cavity)
Feedback: lock frequency to atomic resonance
Frequency division
(electronic counters)
$\div 10^3$
$\div 10^3$
$\div 10^3$
$\div 10^x$
9 192 631 770 Hz → 1 Hz
1 Hz output
(1 pulse per second)
1 s
t

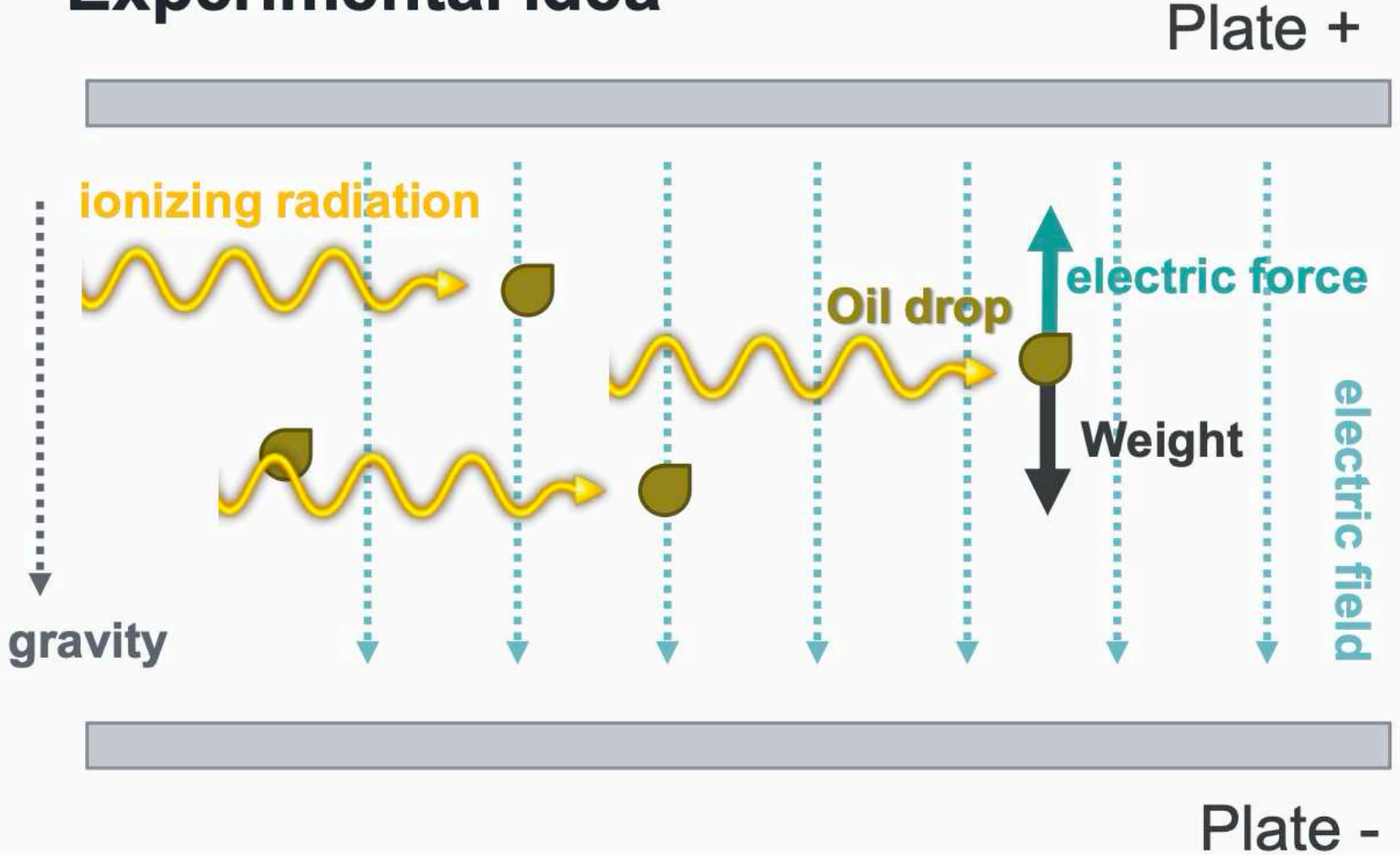

Acc.V Spot Magn Det WD Exp
30.0 kV 3.0 11397x SE 8.0 3
2 µm

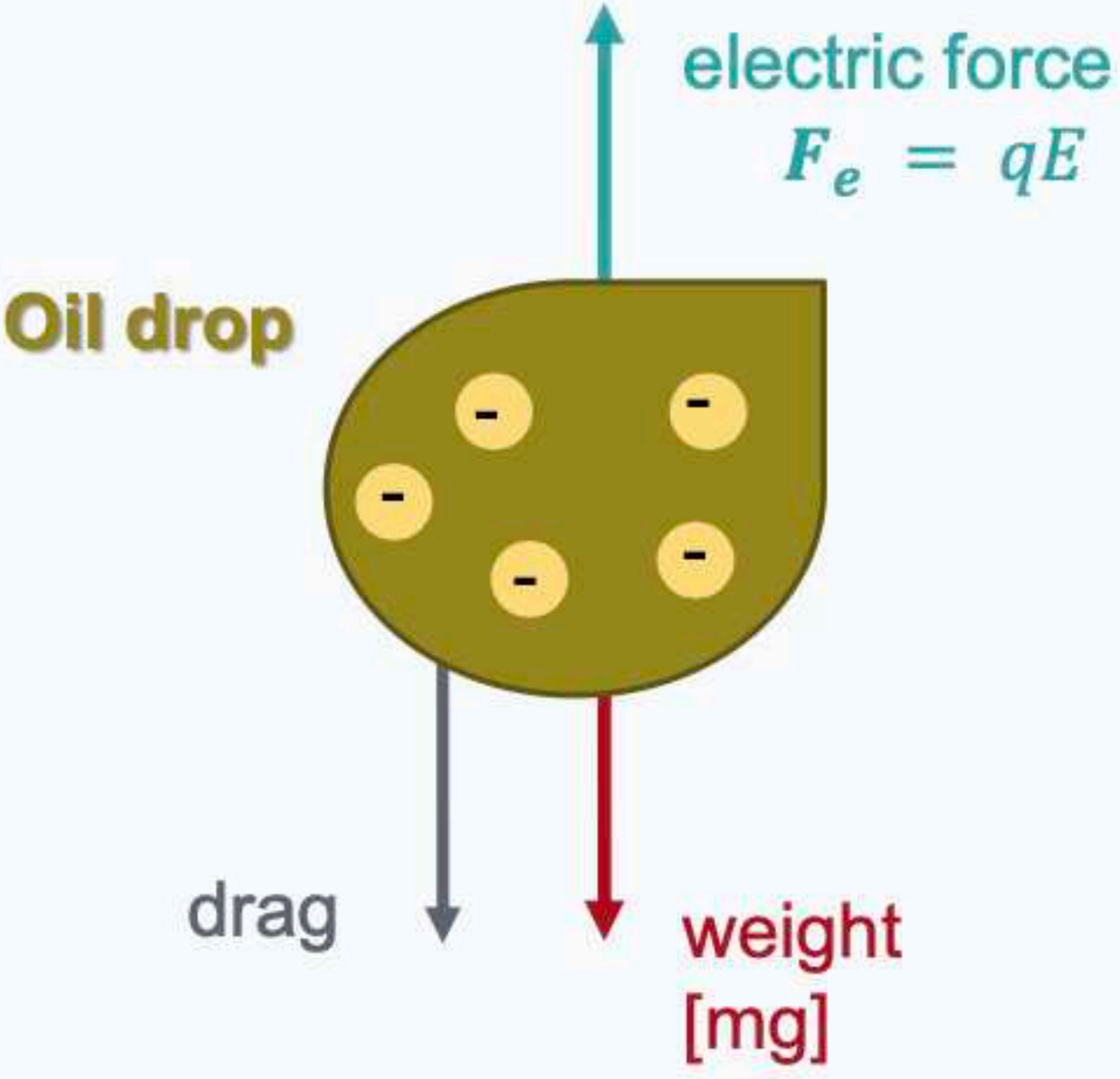

100 nm


Fiera di Padova
Via Niccolò Tommaseo
Via Luigi Piccinato
Via Ugo Bassi
Via Francesco Rismondo
Via San Leonardo Murialdo
Via del Pesca
Via de
Hotel NH Padova
6 min
1,9 km
Liceo Artistico Statale "Amedeo Modigliani"
Centro Copia Stecchini - Tesi 24 ore
Copisteria
Via Venezia
Dipartimento di Matematica "Tullio...
Via Trieste
Via Leonardo Loredan
Best Western Plus Hotel Galileo
Porta Portello
Lungargine del Piovego
Via Giovanni Gradenigo
Dipartimento di Fisica Galileo Galilei
San Gaetano

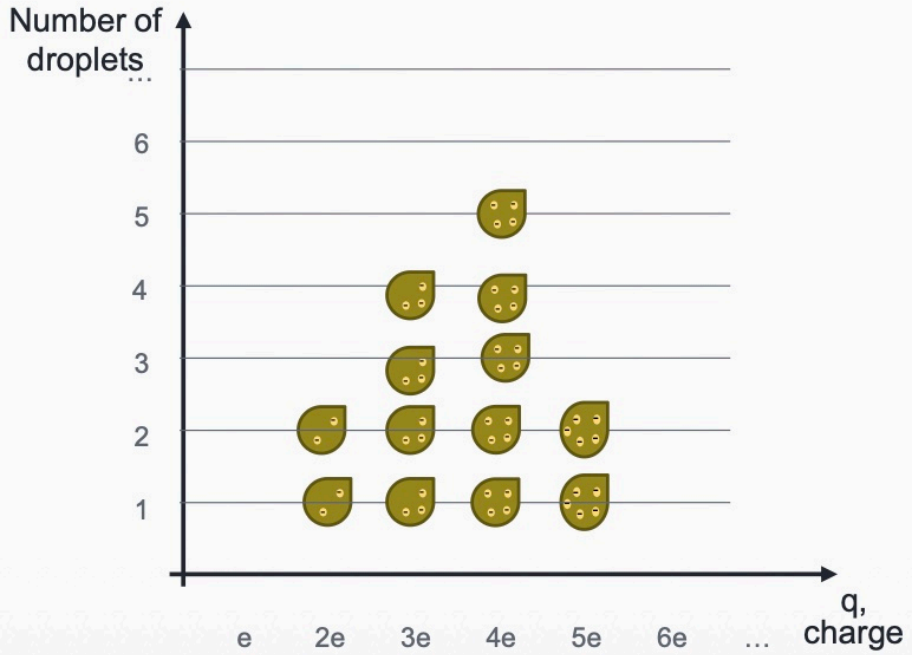


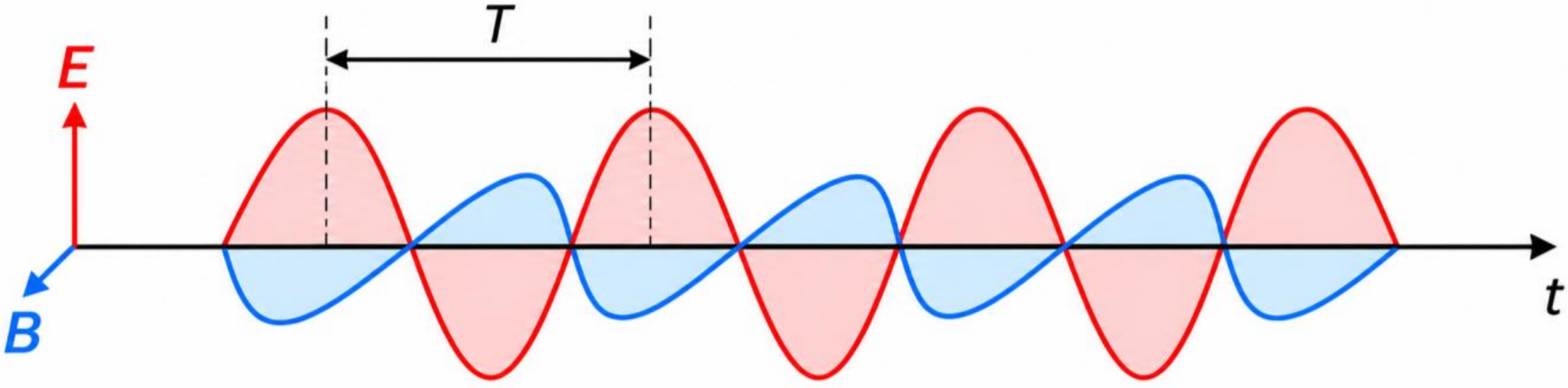
Experimental idea

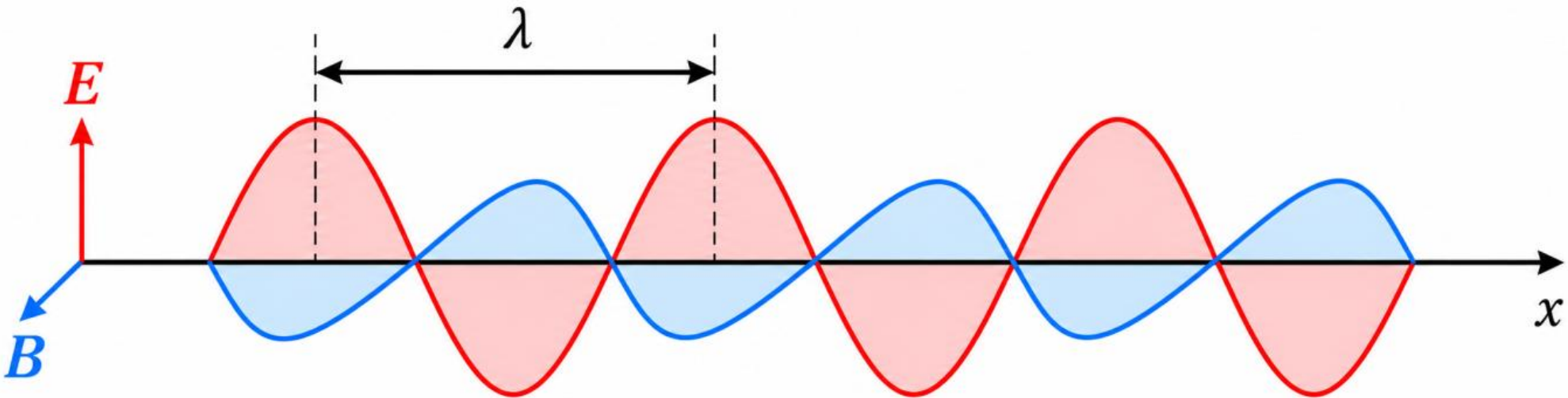


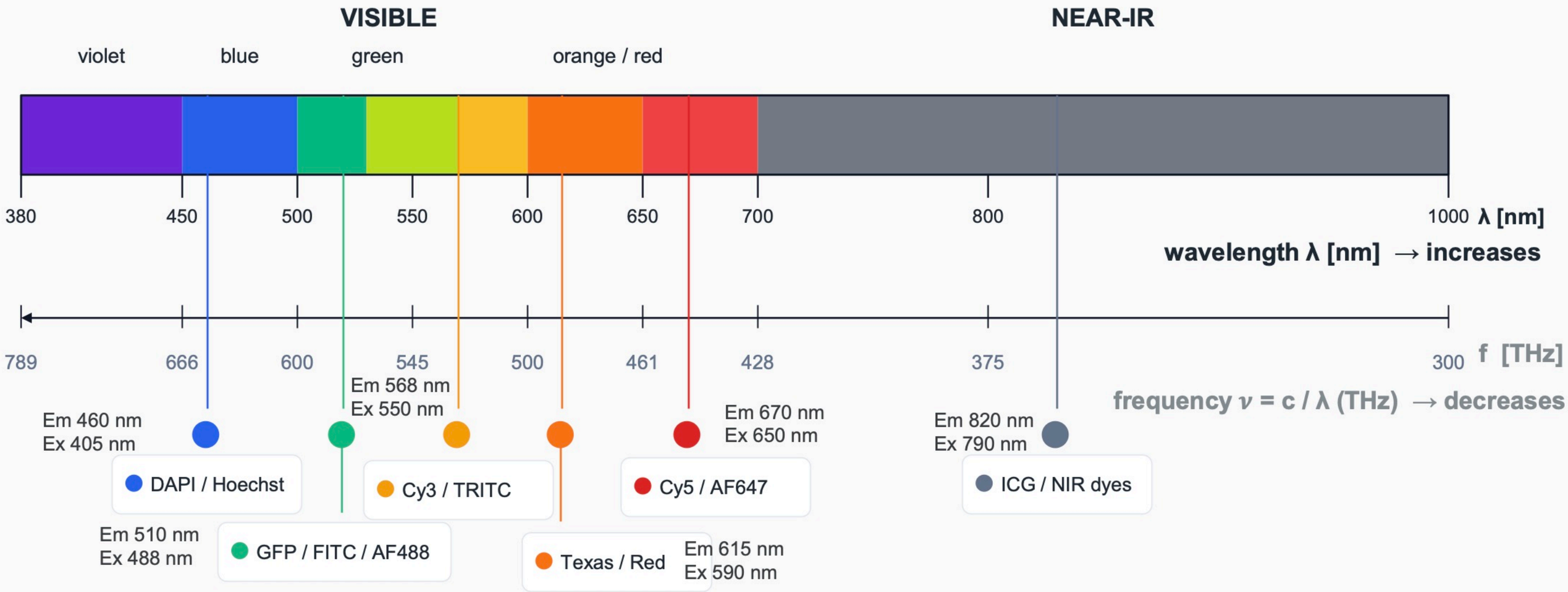

electric force
$F_e = qE$
Oil drop
drag
weight
[mg]

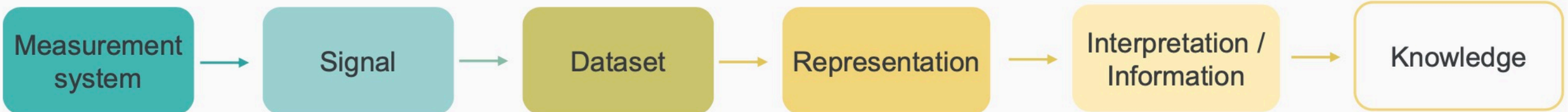

q/e
6
5
4
3
2
1
#1
#2
#3
#4
#5
#6
#7
#8
#9
#10
#11
#12
#13
#14
Droplet #

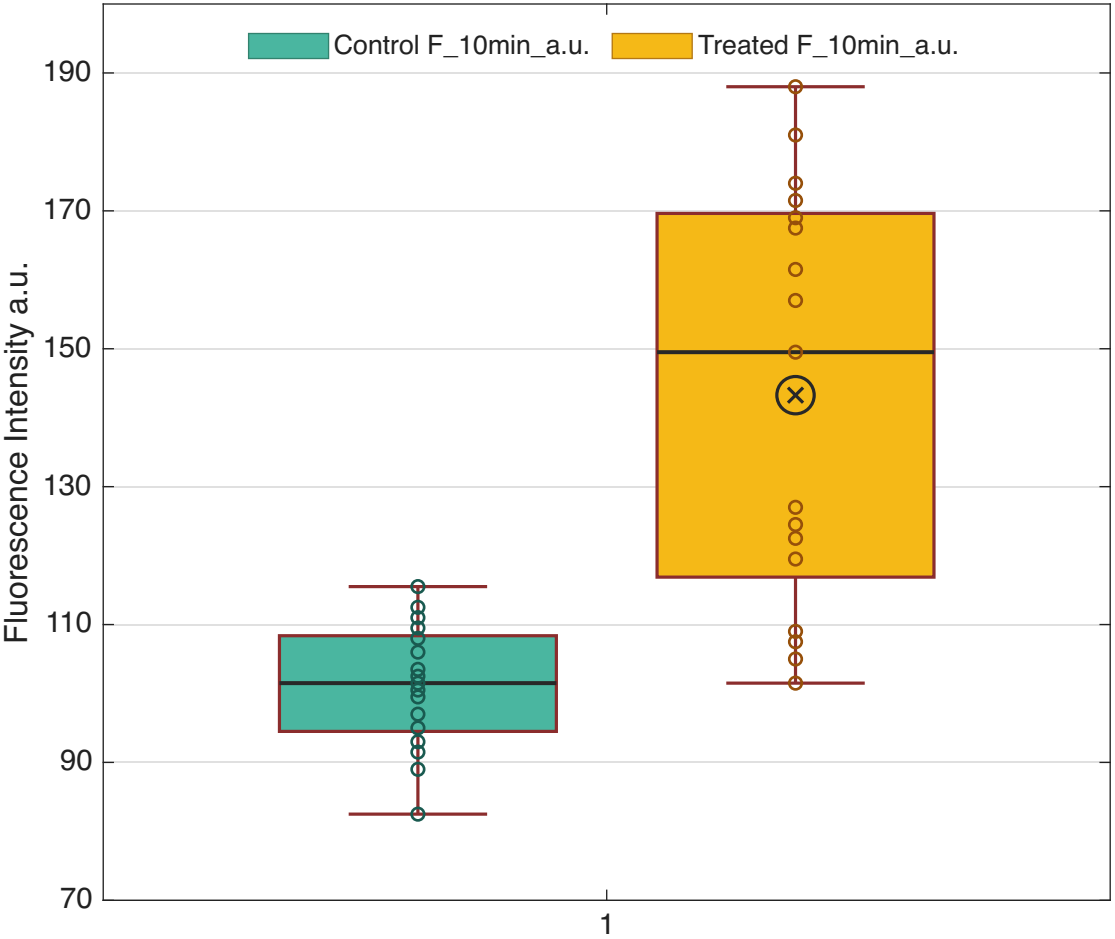

Number of droplets
...
6
5
4
3
2
1
e
2e
3e
4e
5e
6e
...
q, charge

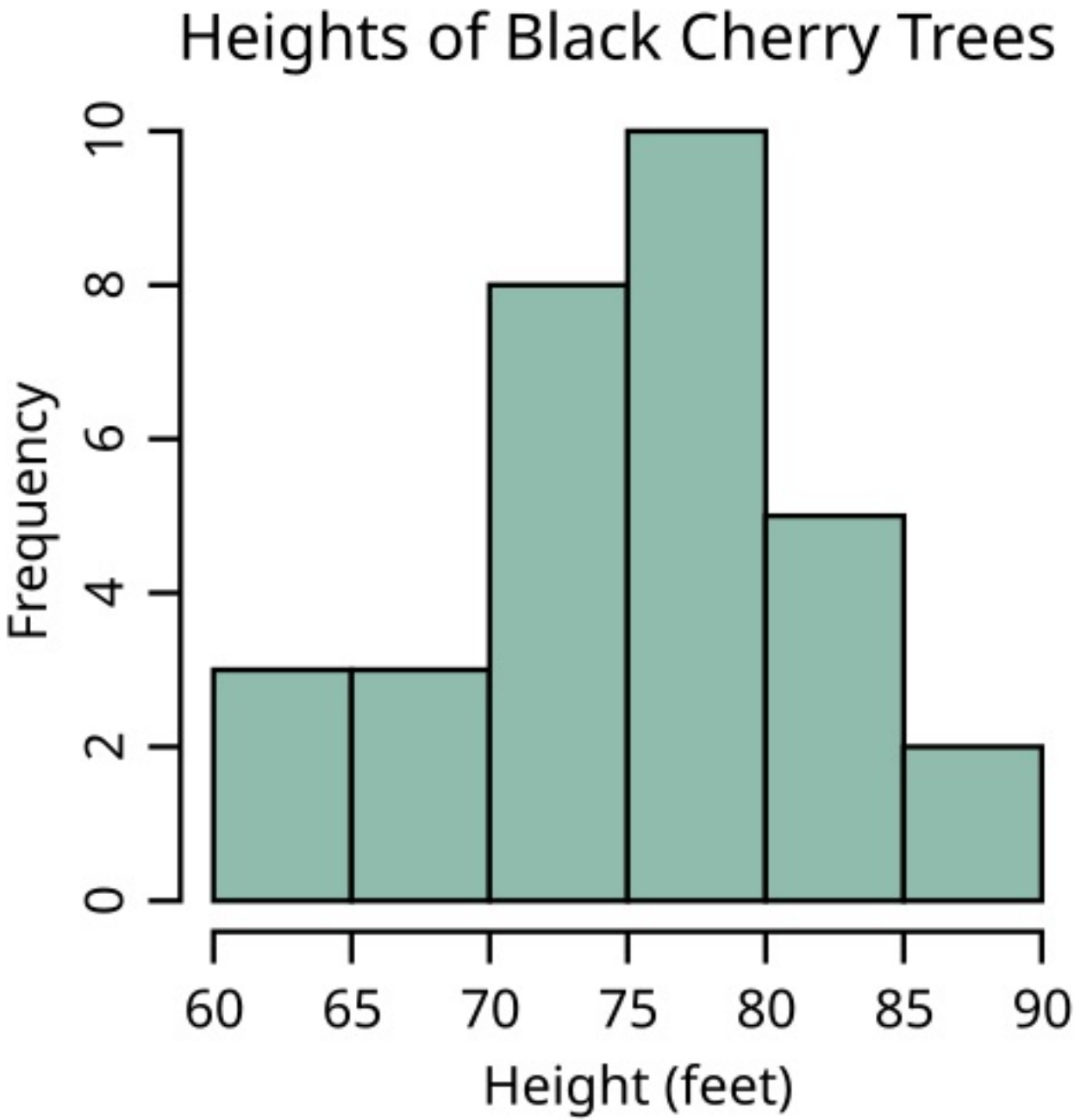

T
E
B
t

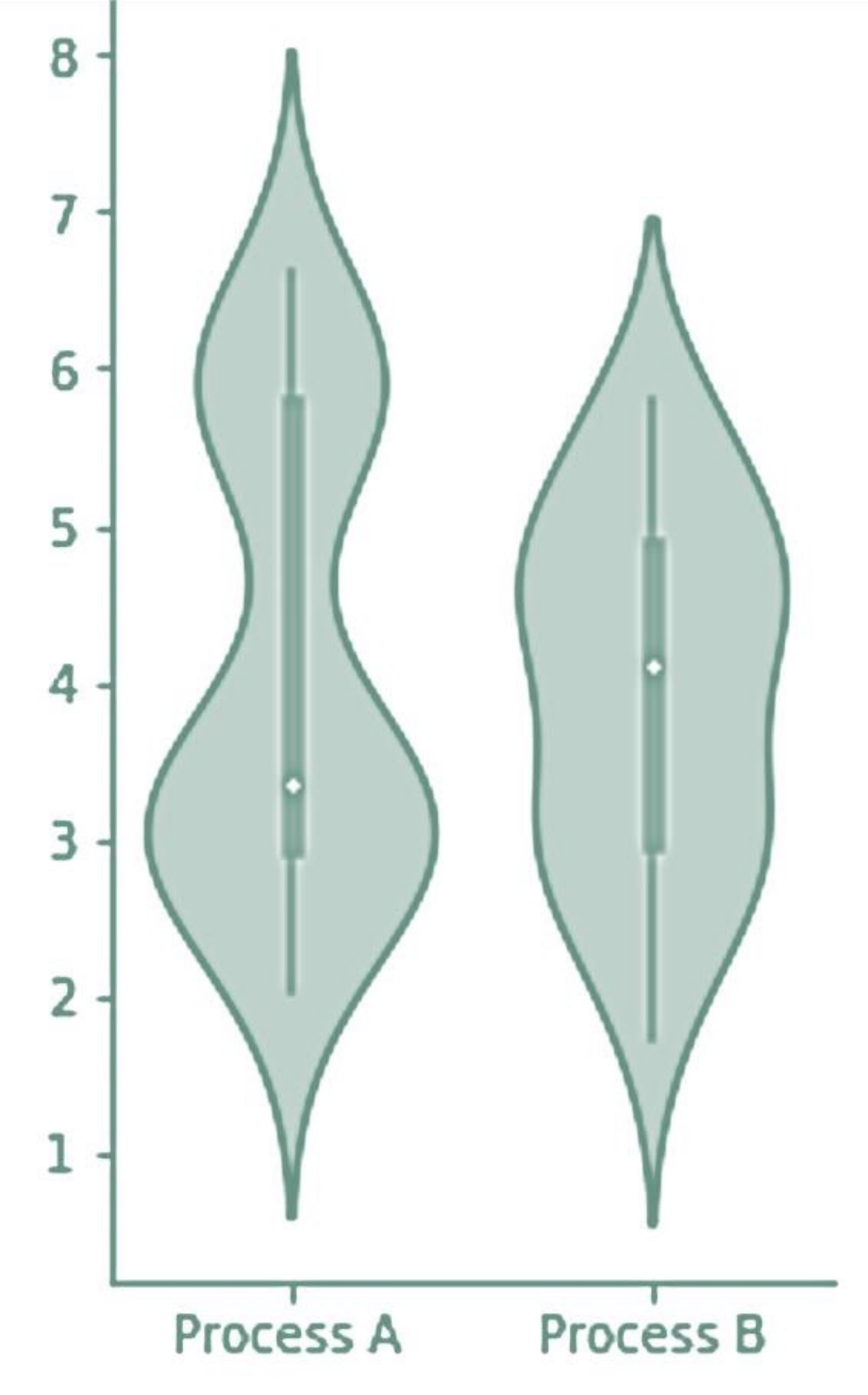

λ
E
B
x

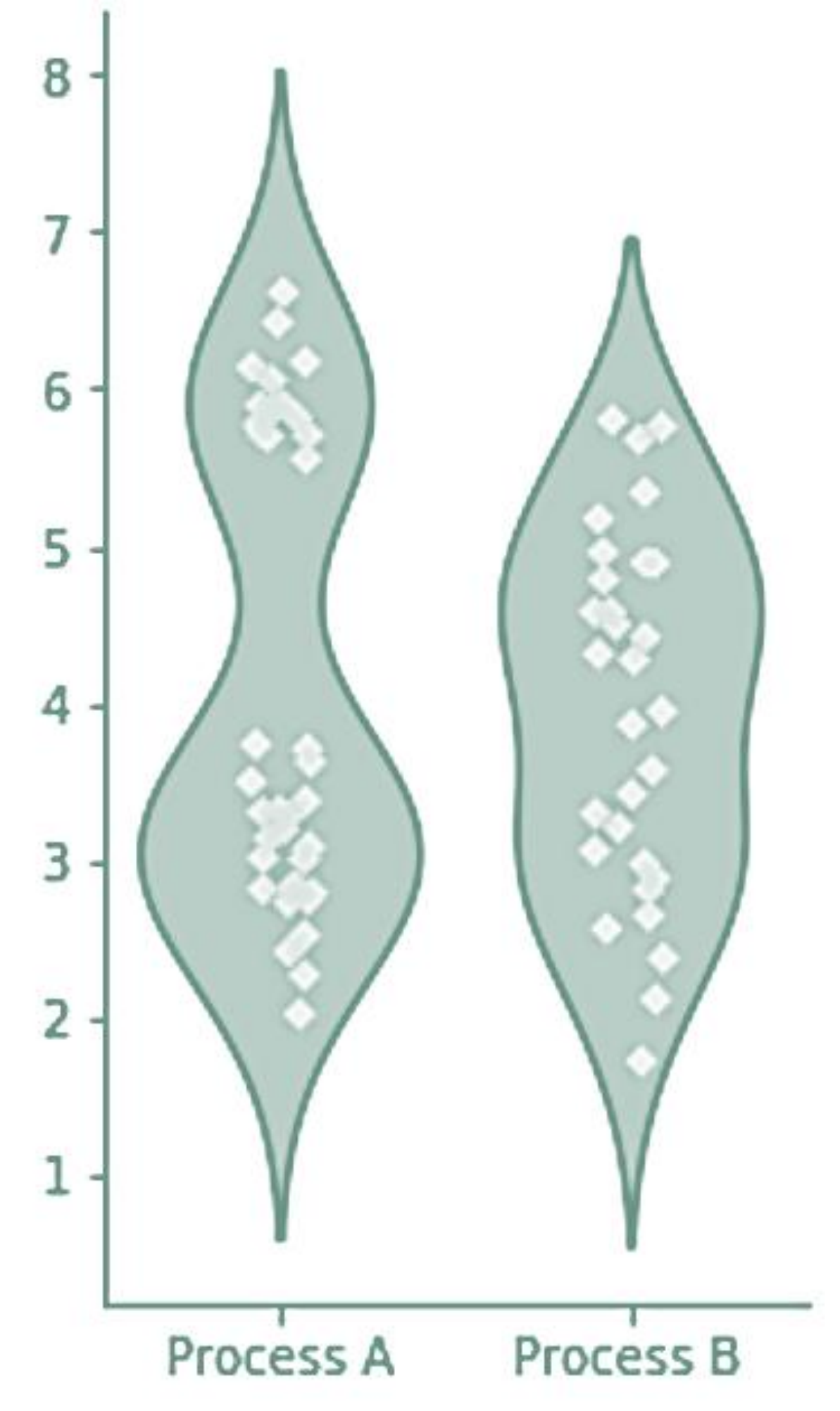

VISIBLE
NEAR-IR
violet
blue
green
orange / red
380
450
500
550
600
650
700
800
1000
λ [nm]
wavelength λ [nm] → increases
789
666
600
545
500
461
428
375
300
f [THz]
frequency ν = c / λ (THz) → decreases
Em 460 nm
Ex 405 nm
DAPI / Hoechst
Em 510 nm
Ex 488 nm
GFP / FITC / AF488
Em 568 nm
Ex 550 nm
Cy3 / TRITC
Texas / Red
Em 615 nm
Ex 590 nm
Em 670 nm
Ex 650 nm
Cy5 / AF647
Em 820 nm
Ex 790 nm
ICG / NIR dyes

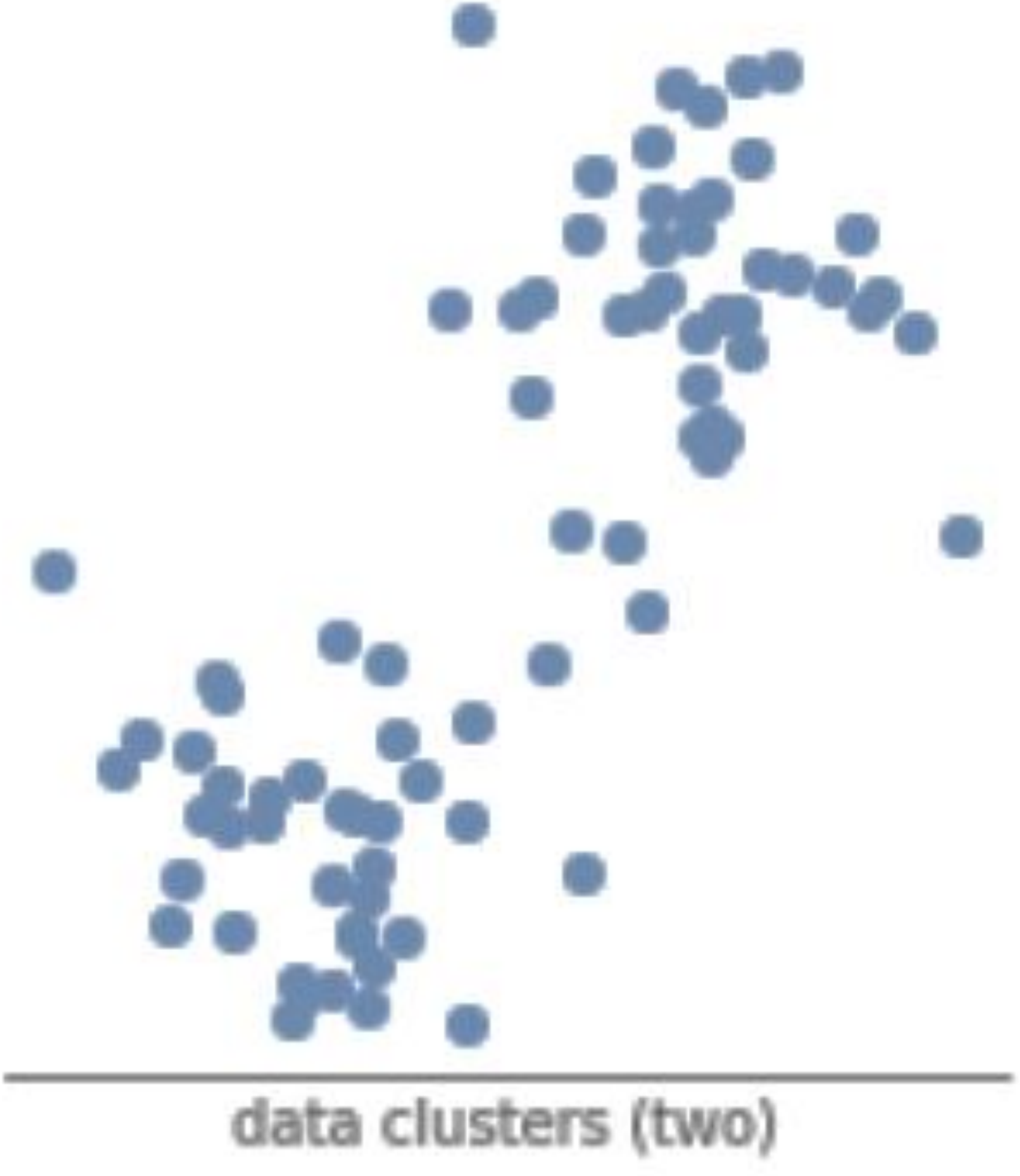

Measurement system
Signal
Dataset
Representation
Interpretation / Information
Knowledge

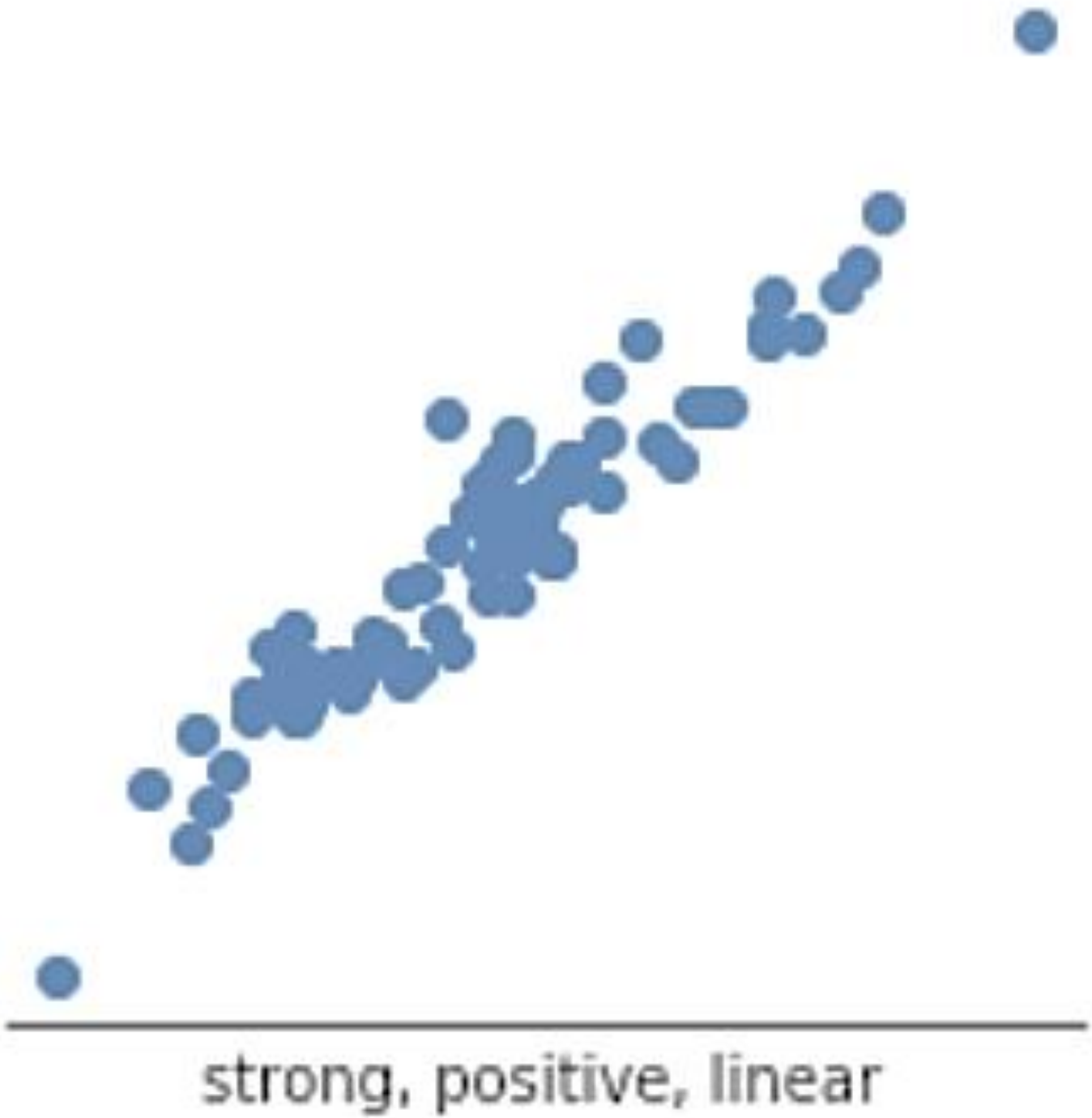

Control F_10min_a.u.
Treated F_10min_a.u.
Fluorescence Intensity a.u.
190
170
150
130
110
90
70
1

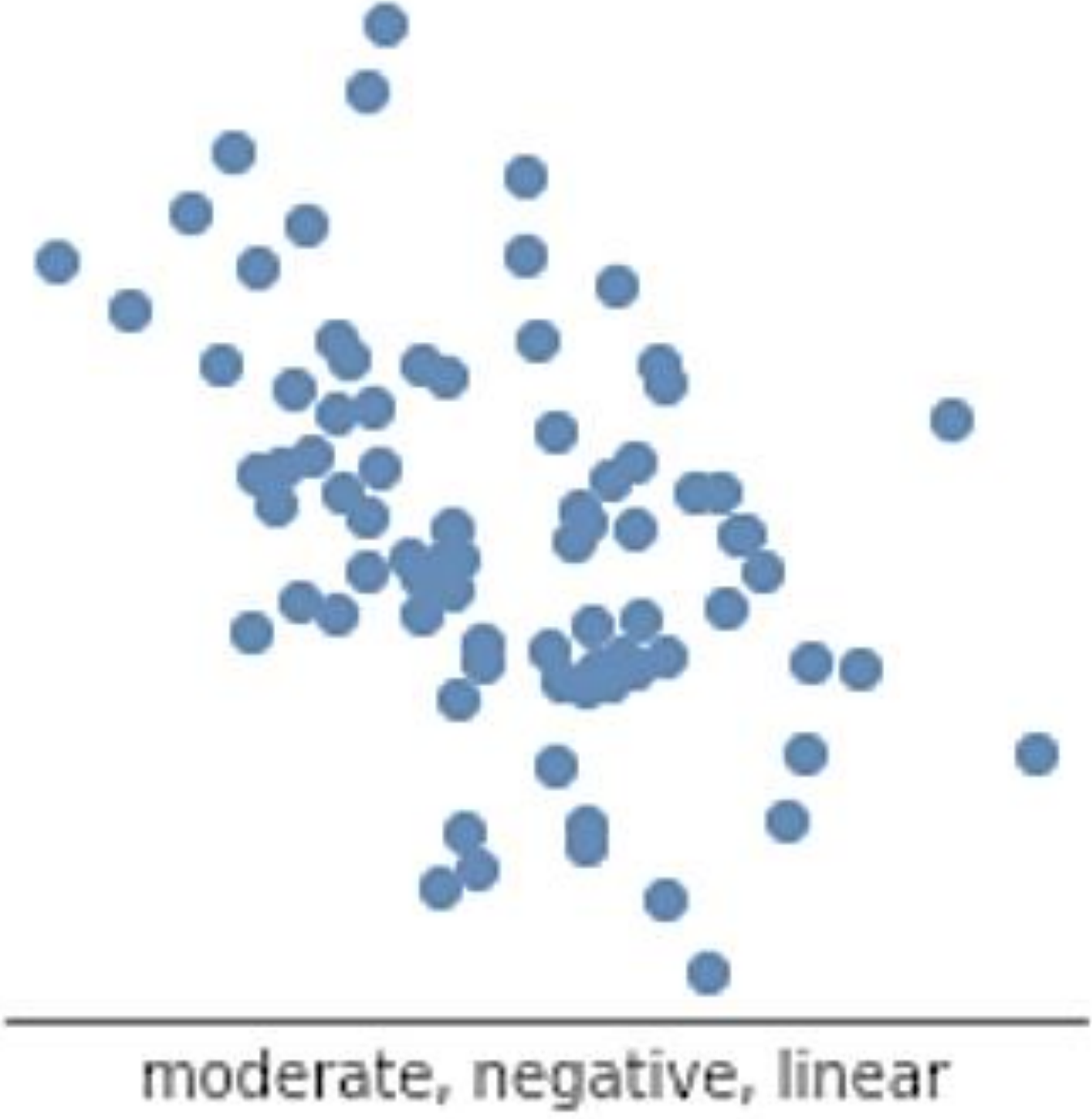

Heights of Black Cherry Trees
Frequency
0
2
4
6
8
10
60
65
70
75
80
85
90
Height (feet)

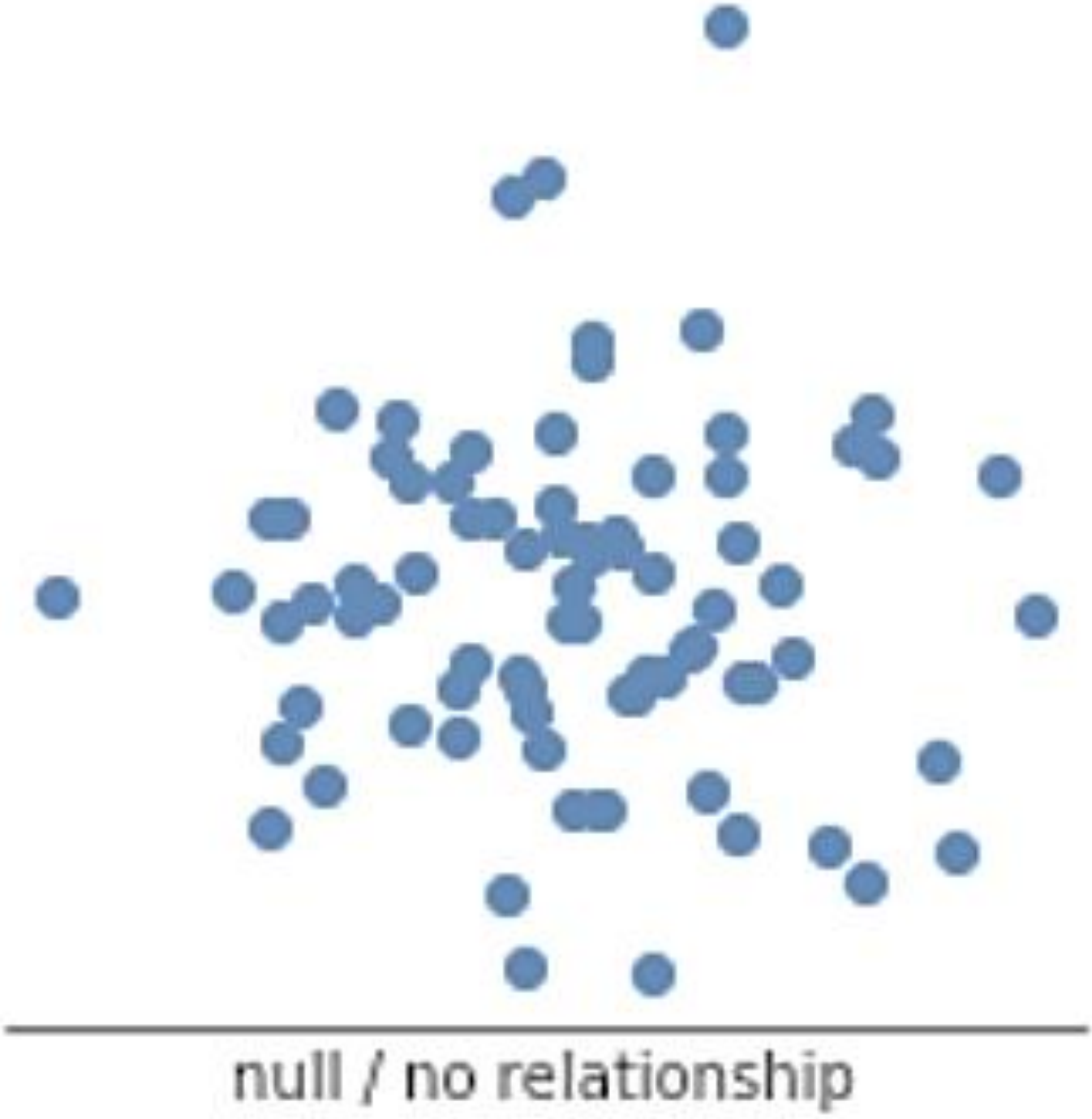

8
7
6
5
4
3
2
1
Process A
Process B

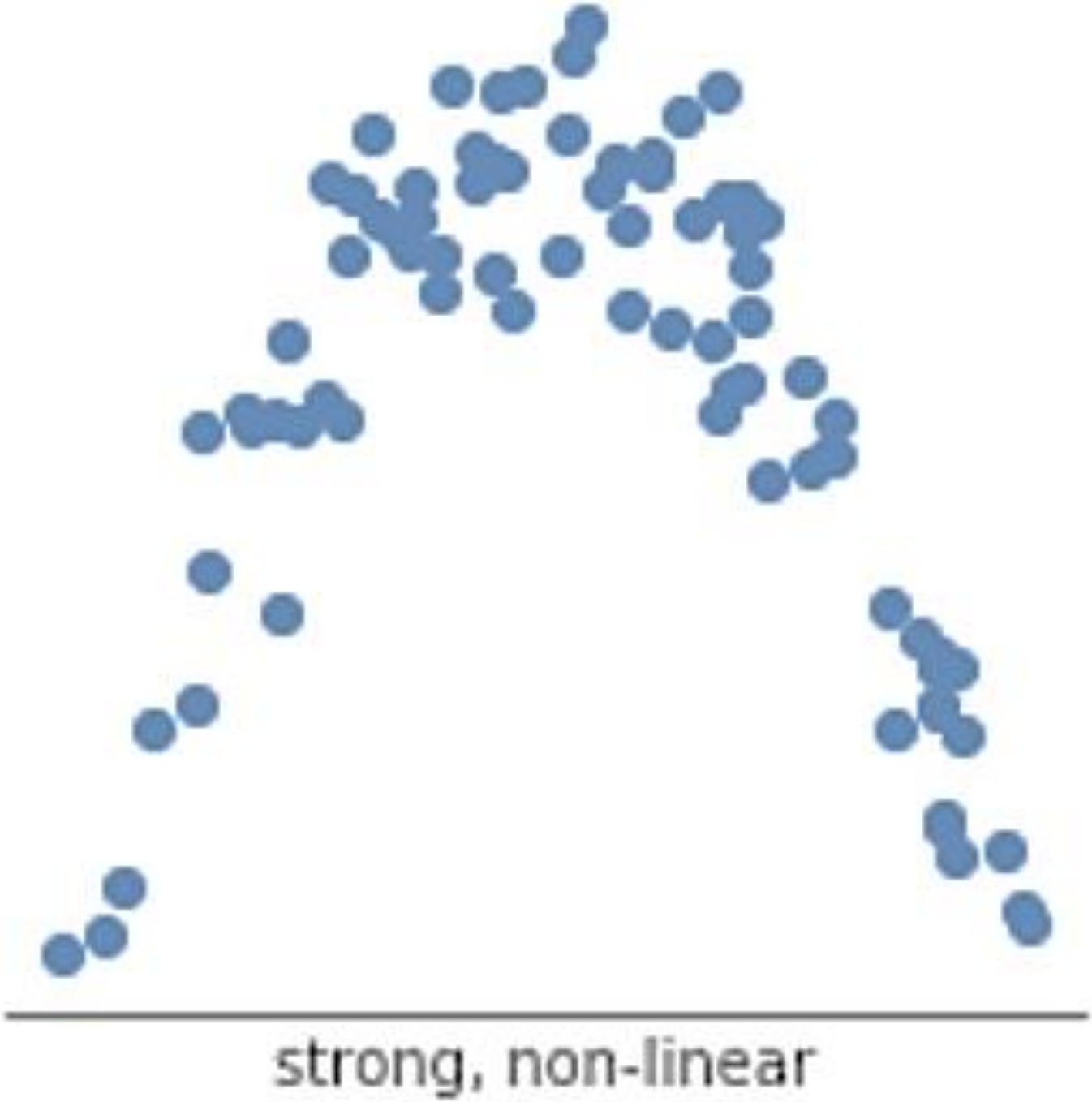

8
7
6
5
4
3
2
1
Process A
Process B


data clusters (two)

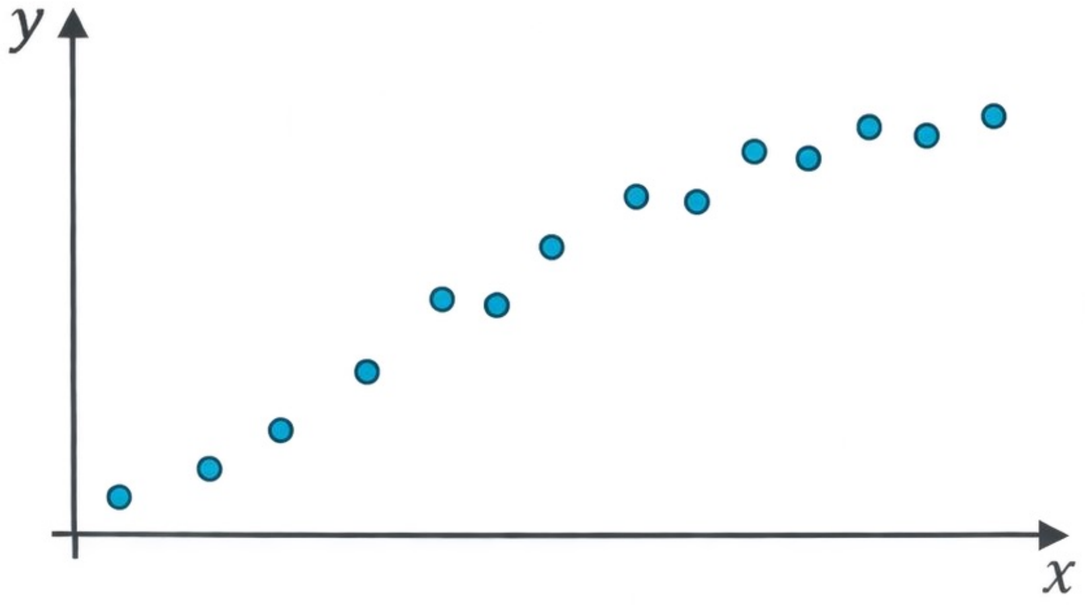

strong, positive, linear


moderate, negative, linear

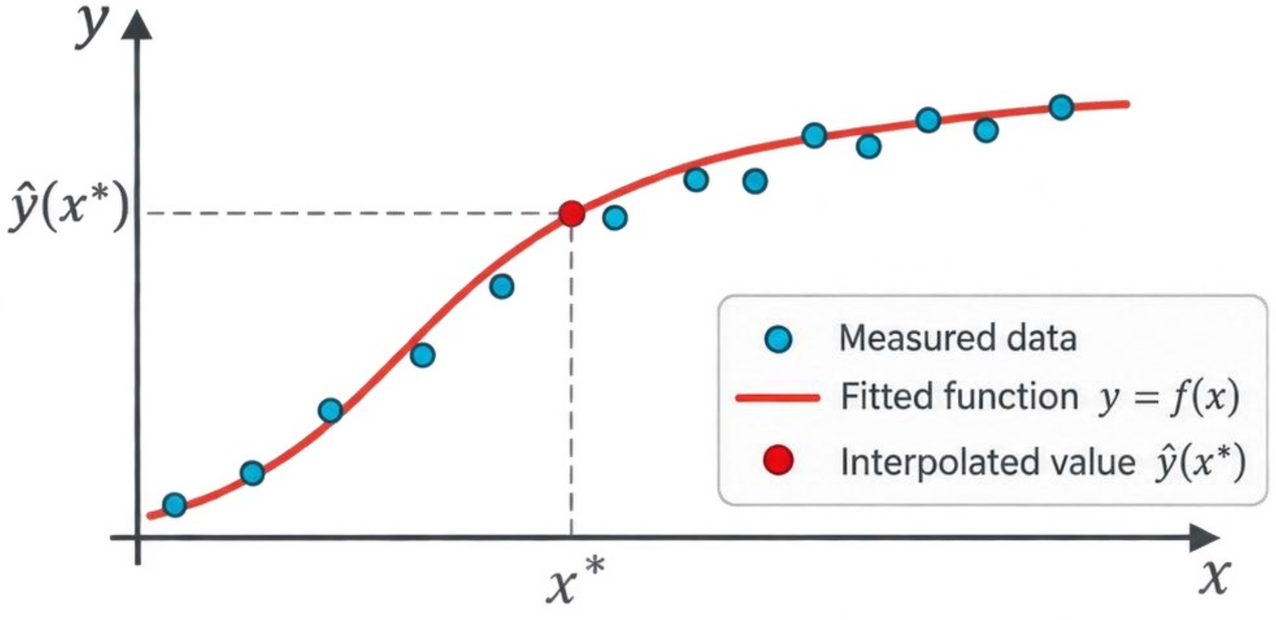

null / no relationship


strong, non-linear

Measured data



Functional approximation

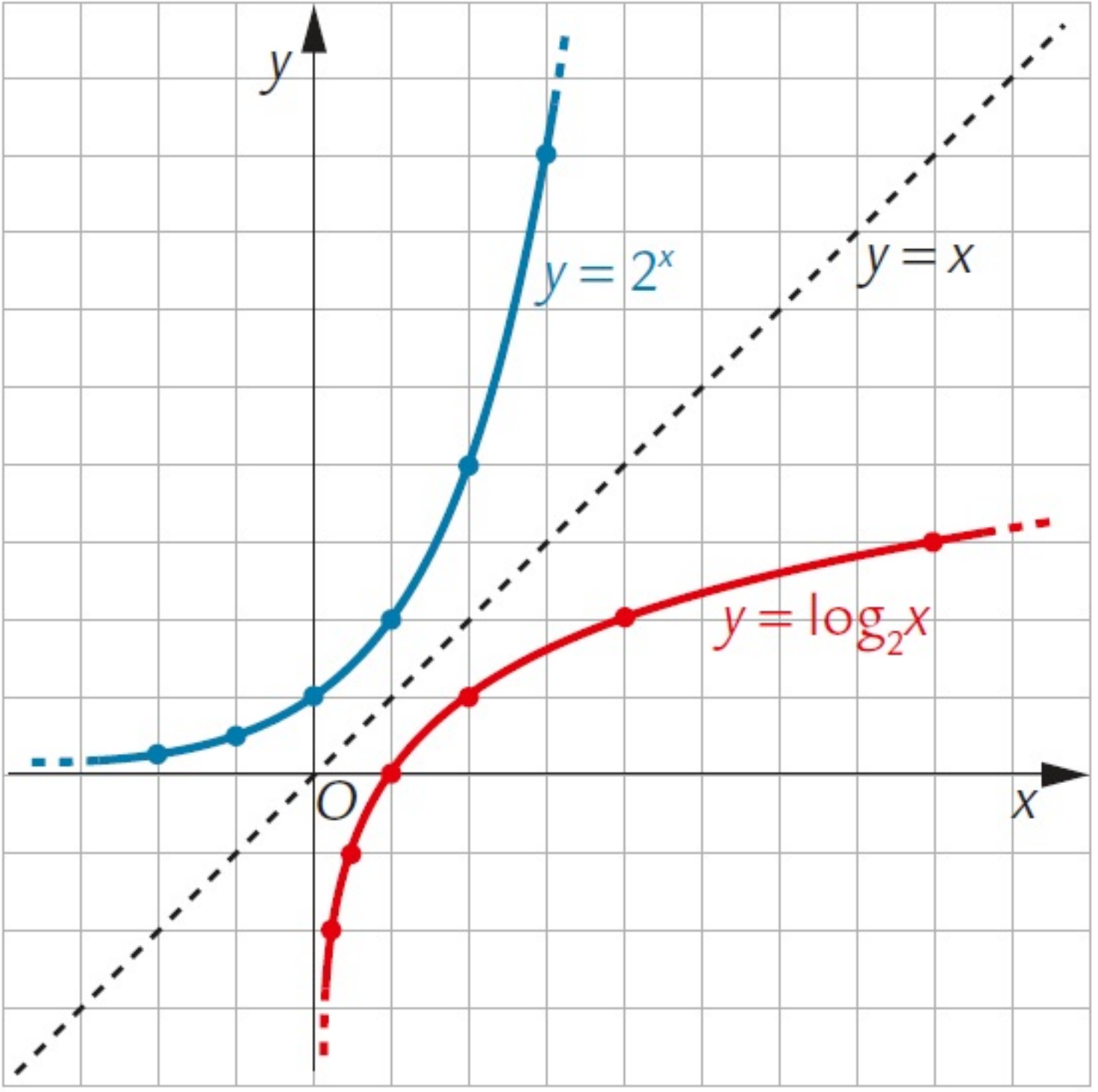


Discrete measurements

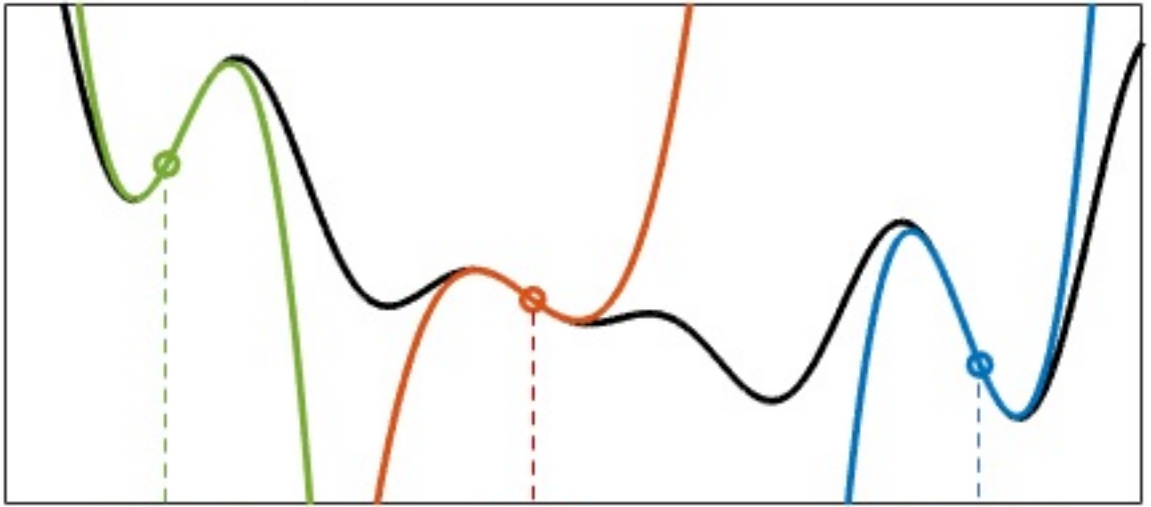
Values of y measured only at specific input points x.

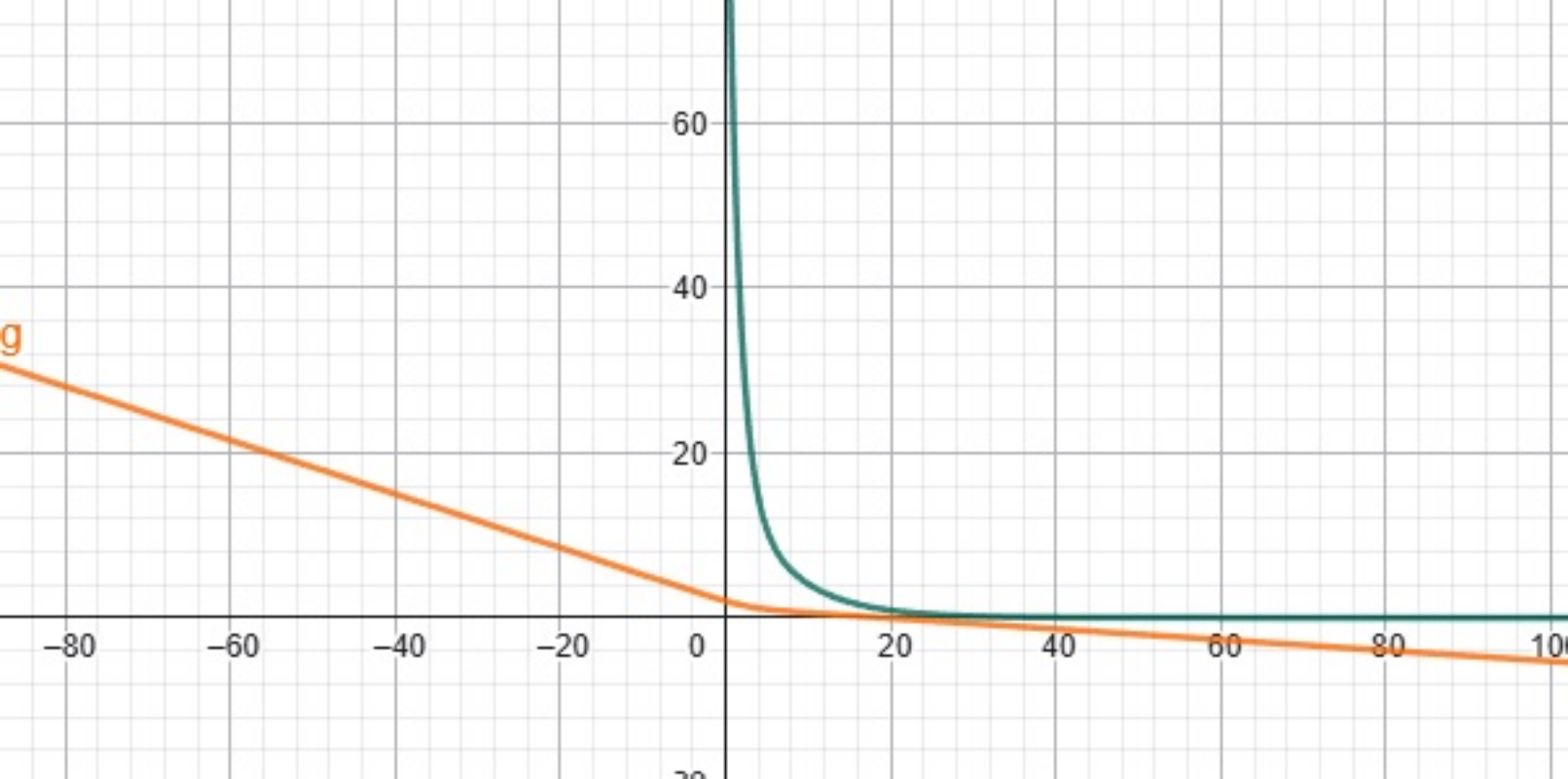
Continuous representation and interpolation

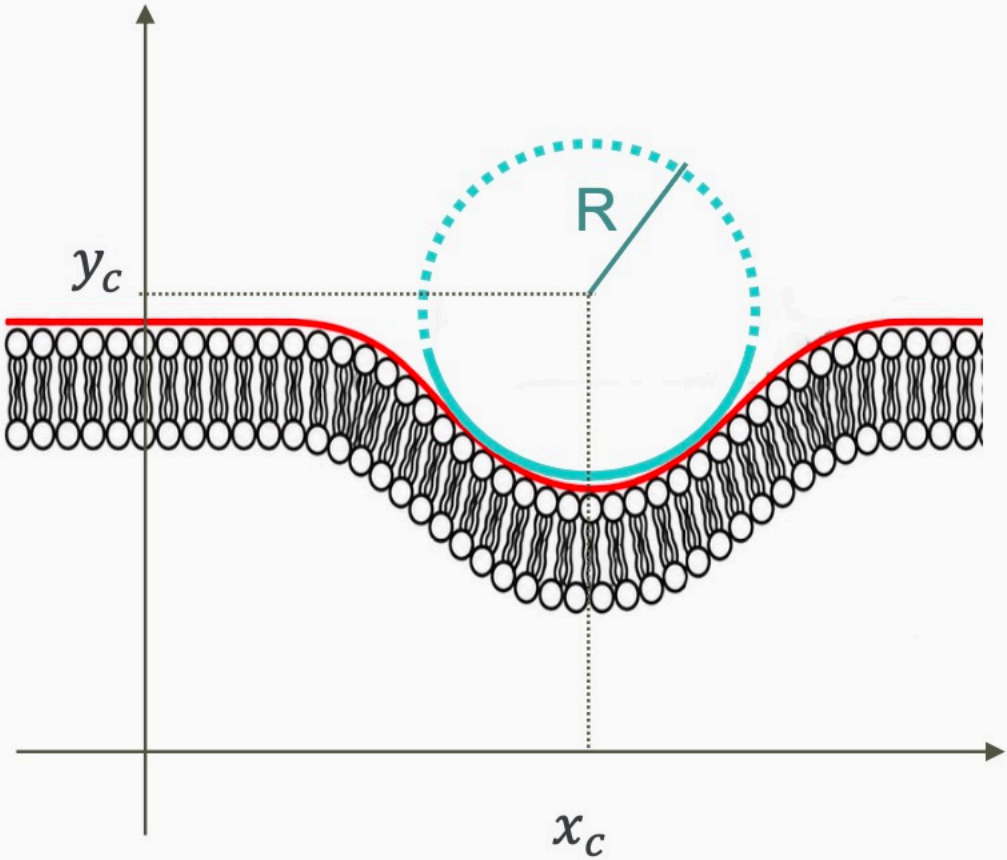
A mathematical function $f(x)$ approximates the data, allowing us to estimate values at unmeasured input points.

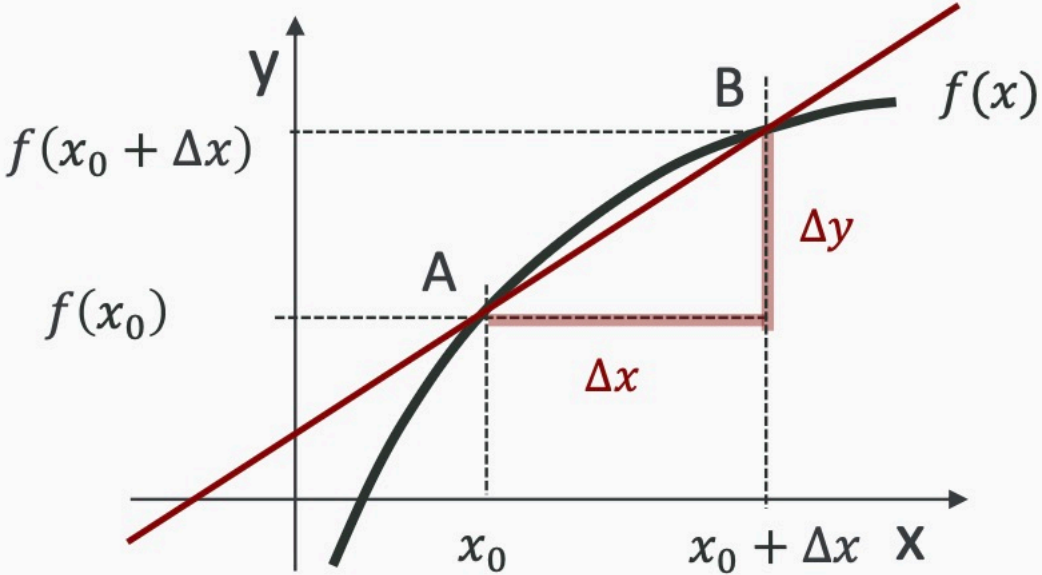

y
x
O
$y = 2^x$
$y = x$
$y = \log_2 x$

Examples with Various Reference Points

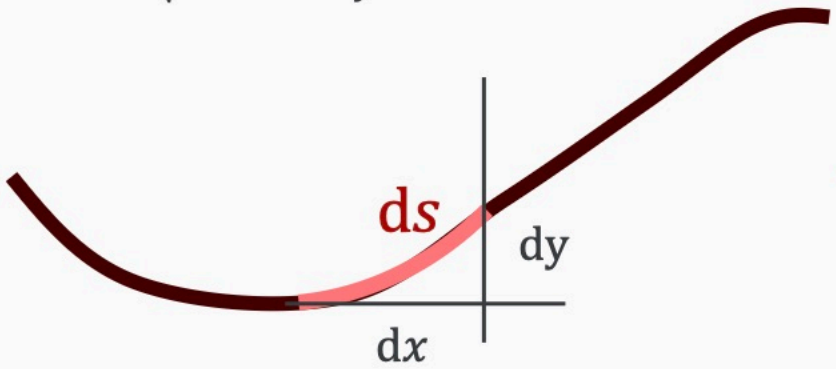


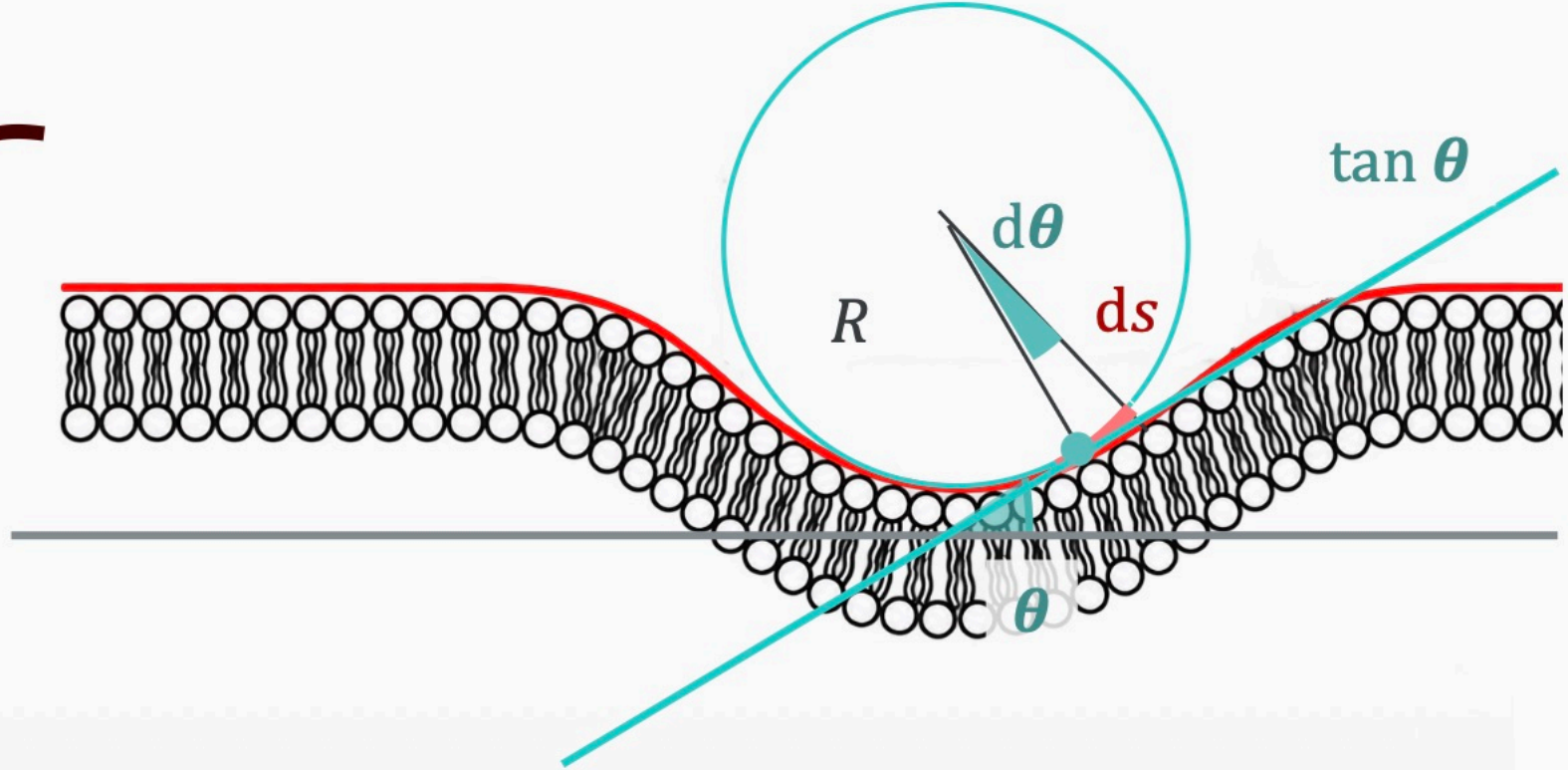

g
60
40
20
−80
−60
−40
−20
0
20
40
60
80

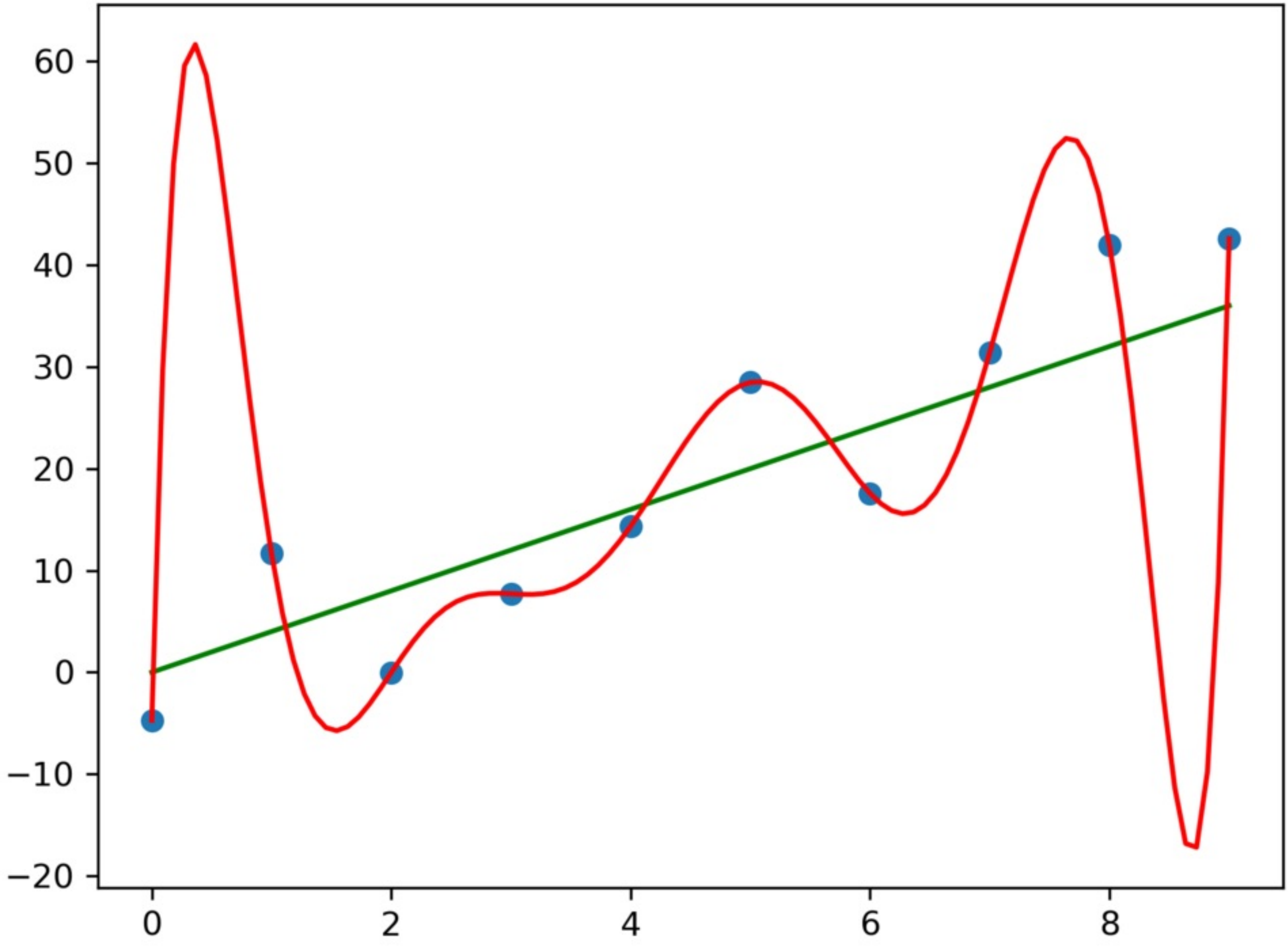

R
y_c
x_c

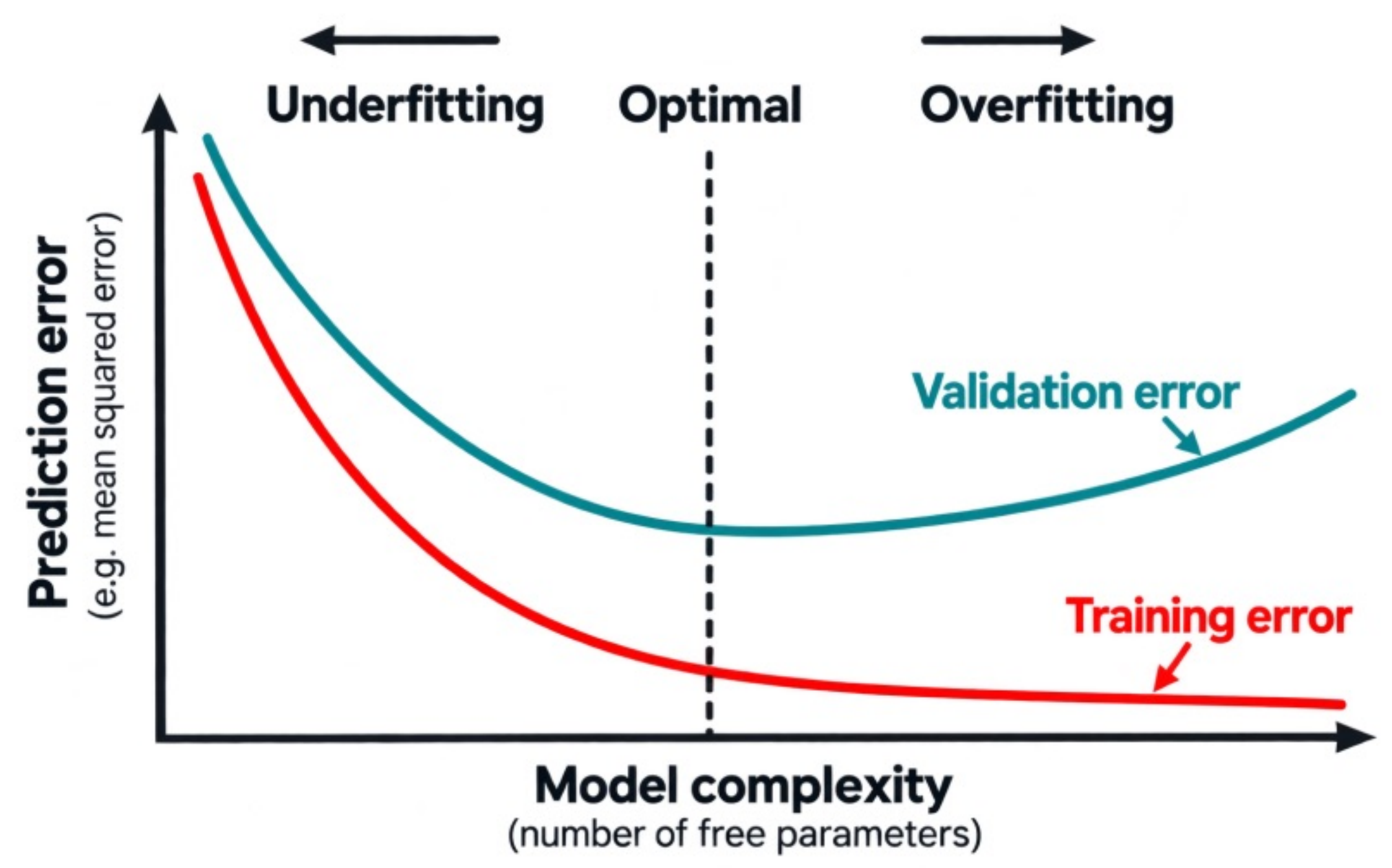

y
B
$f(x)$
$f(x_0 + \Delta x)$
Δy
A
$f(x_0)$
Δx
x_0
$x_0 + \Delta x$
x

$$ds = \sqrt{dx^2 + dy^2}$$






60
50
40
30
20
10
0
−10
−20
0
2
4
6
8


Underfitting
Optimal
Overfitting
Prediction error
(e.g. mean squared error)
Validation error
Training error
Model complexity
(number of free parameters)

Transducer

Input

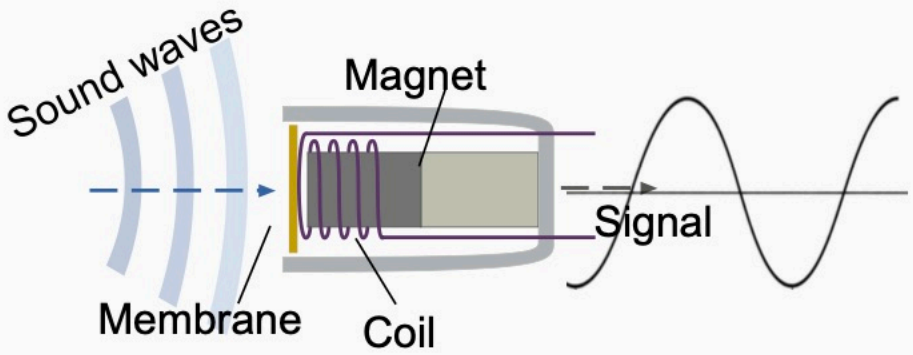
Output

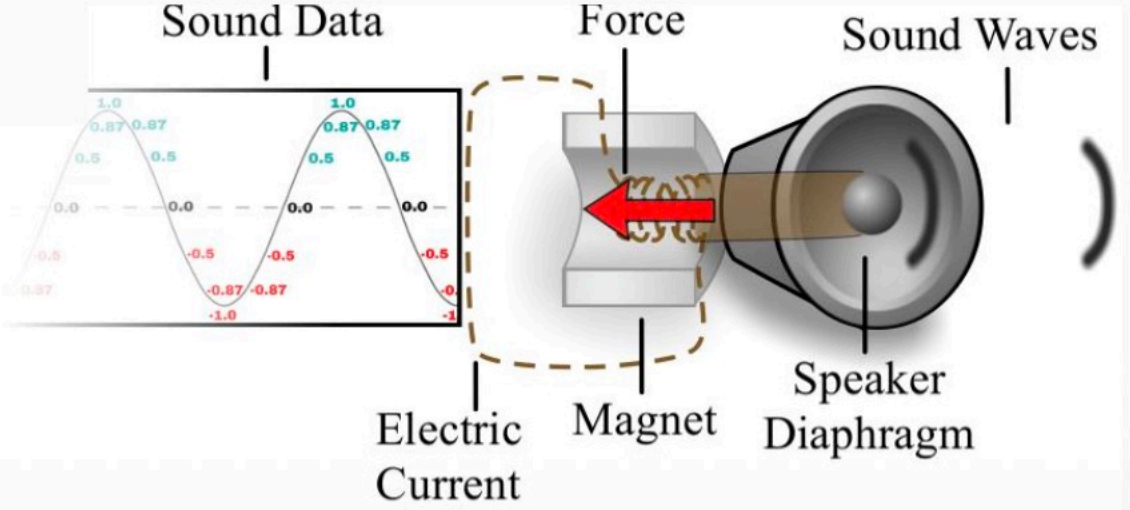
Microphone converts physical sound waves (pressure) into electrical signals (voltage or current).
It acts as an input device to detect sound

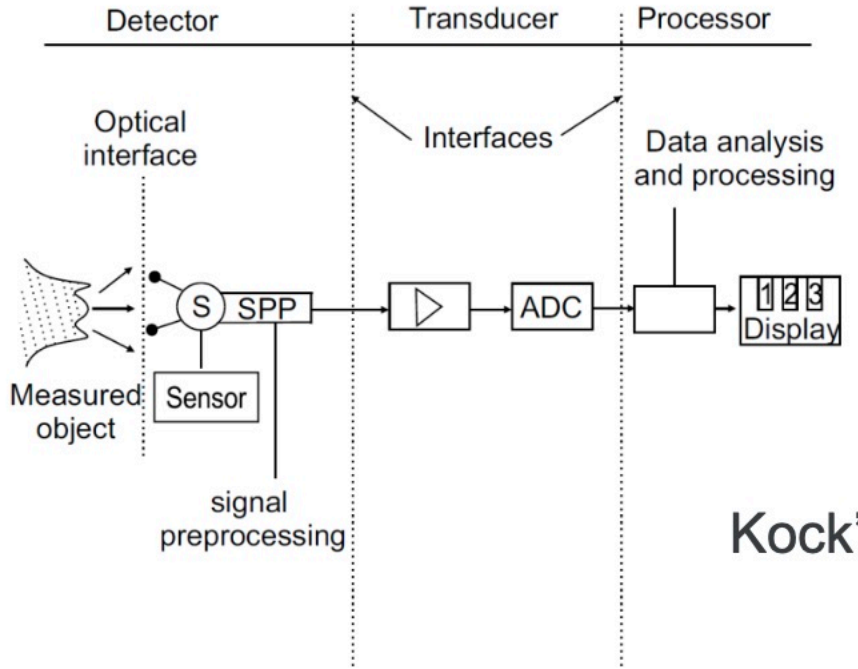
Sensor

Actuator

Speaker is an actuator because it converts electrical signal into physical, mechanical motion







Kock's book

Fig. 7.2 Measurement chain having an optical input interface

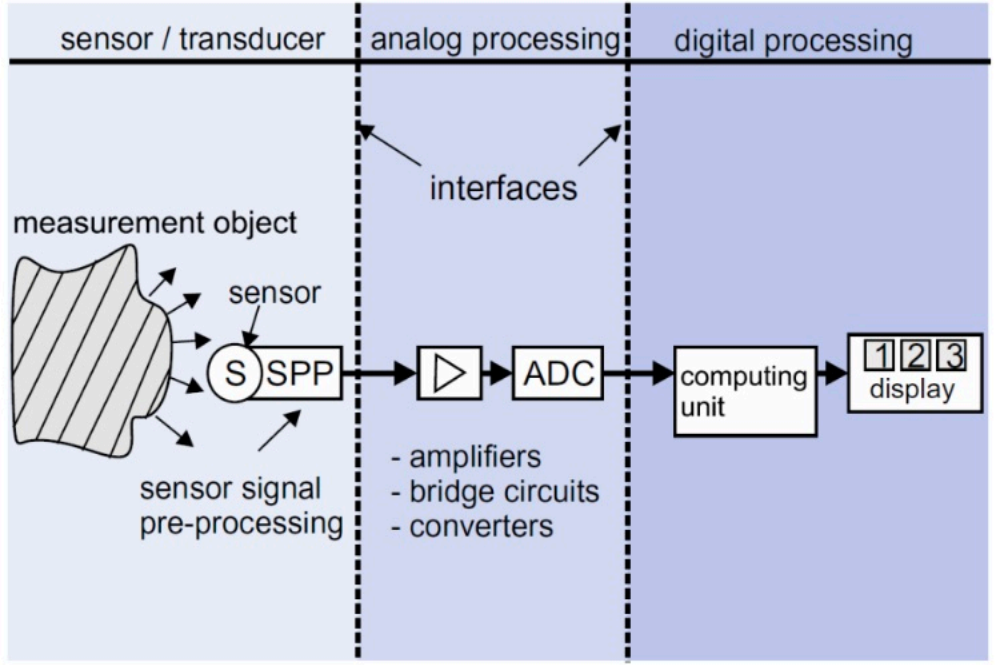


Fig. 1.6 Measurement chain; SPP: signal preprocessing, ADC: analog-to-digital converter

oximate the sensor response.

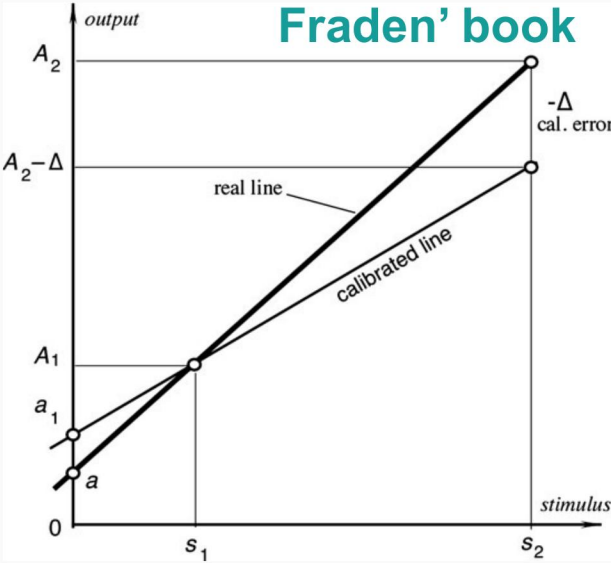
red sensor outputs are subject to
ncorrect estimates of the transfer

del must adequately describe the
. An unsuitable model may

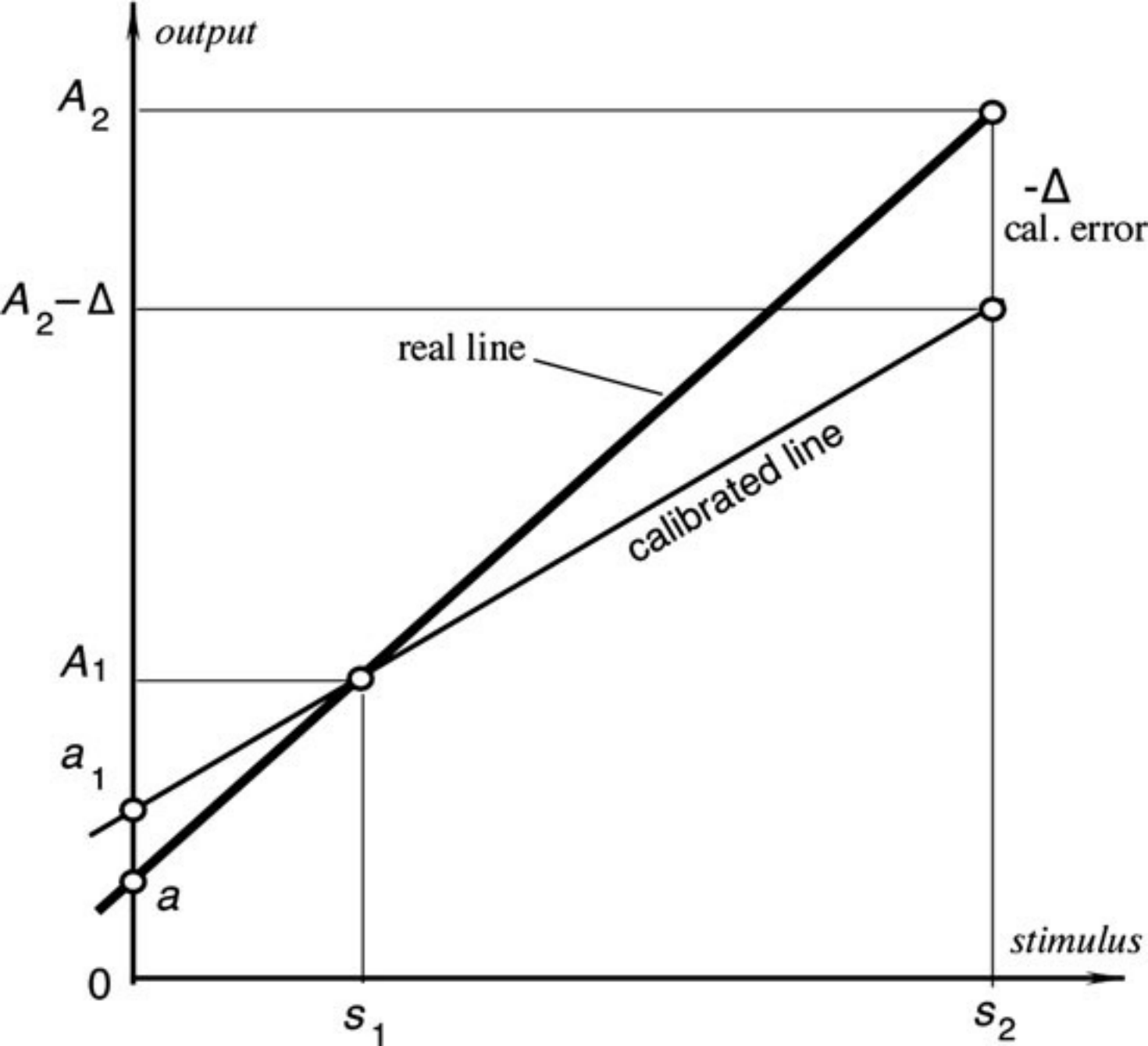
d within specified input ranges
, and sensor drift may alter the

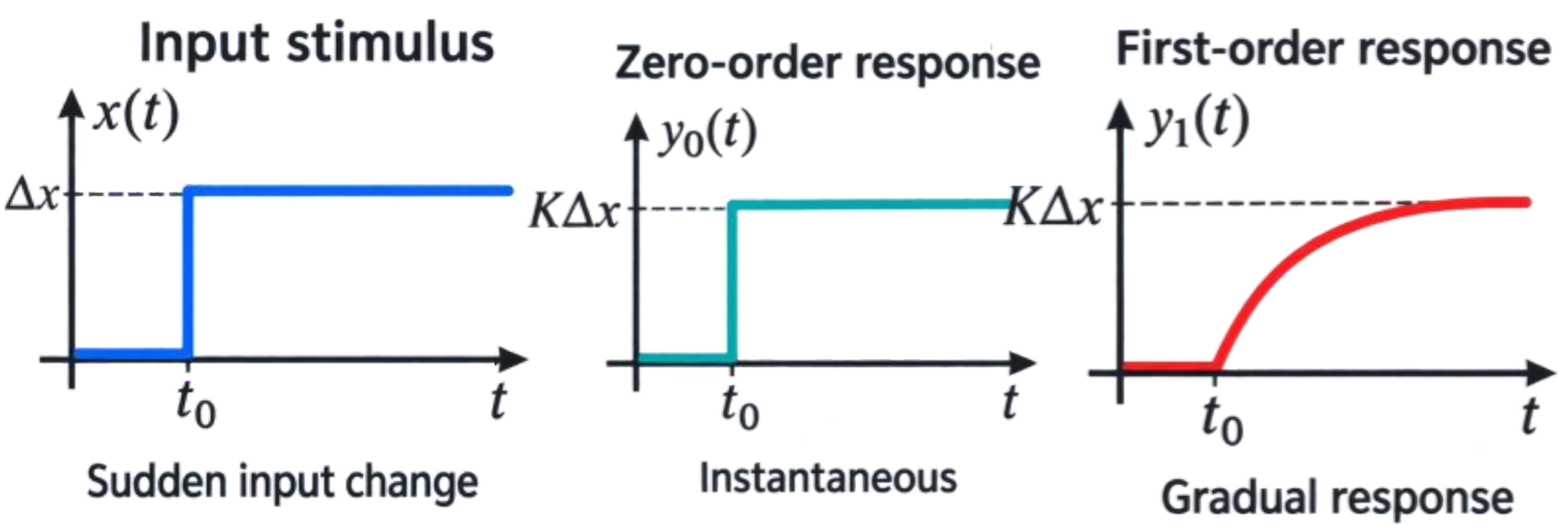
function and therefore into subsequent measurements of unknown inputs.

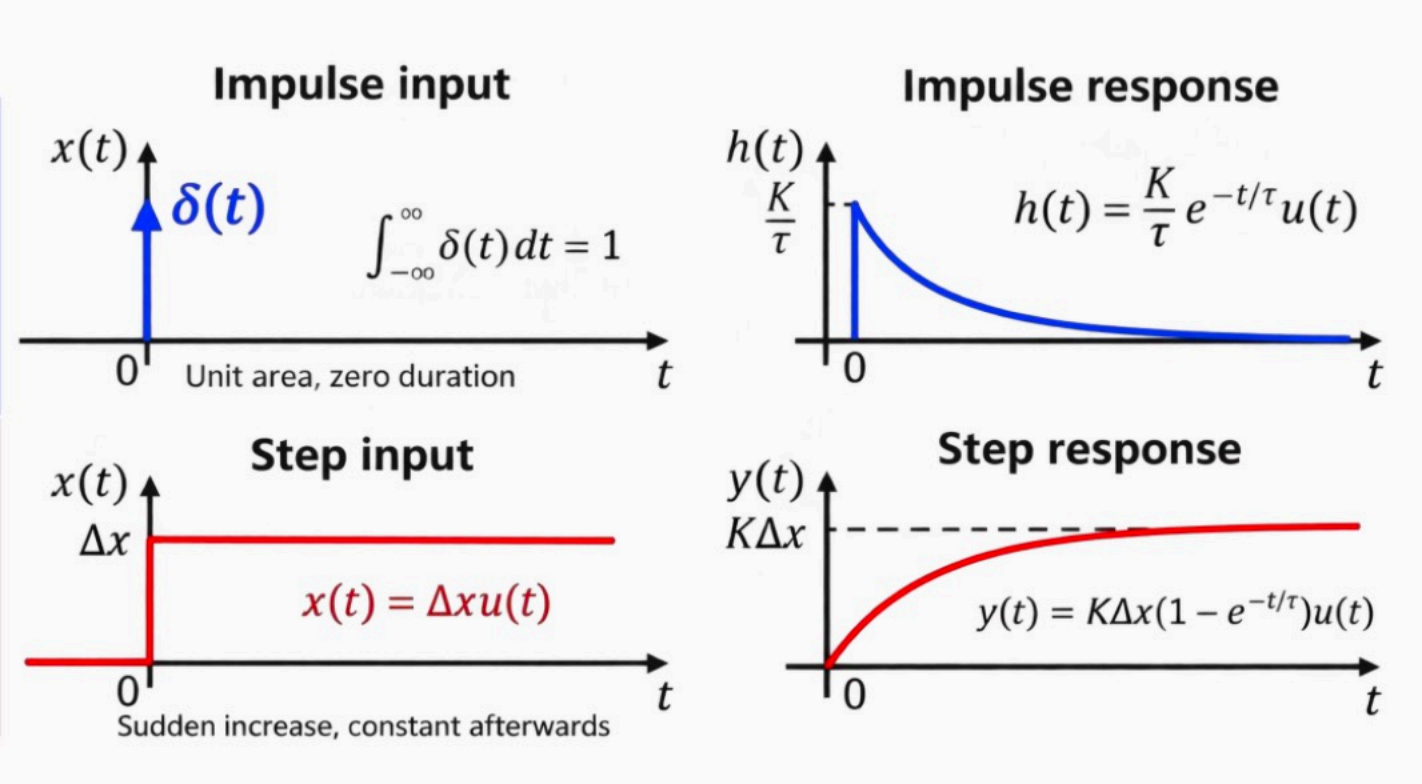
Fraden' book

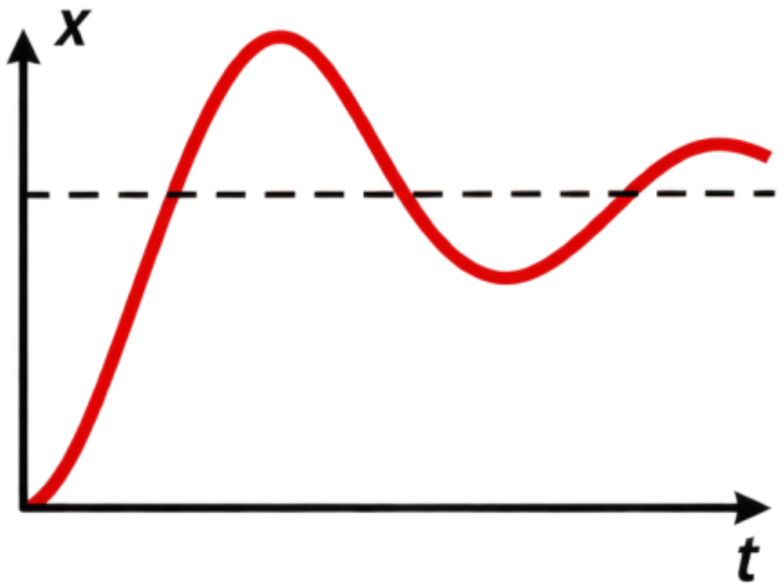


hile the thinner line represents the incorrectly calibrated response. The first
error Δ. As a consequence, the fitted calibration line has an incorrect slope and

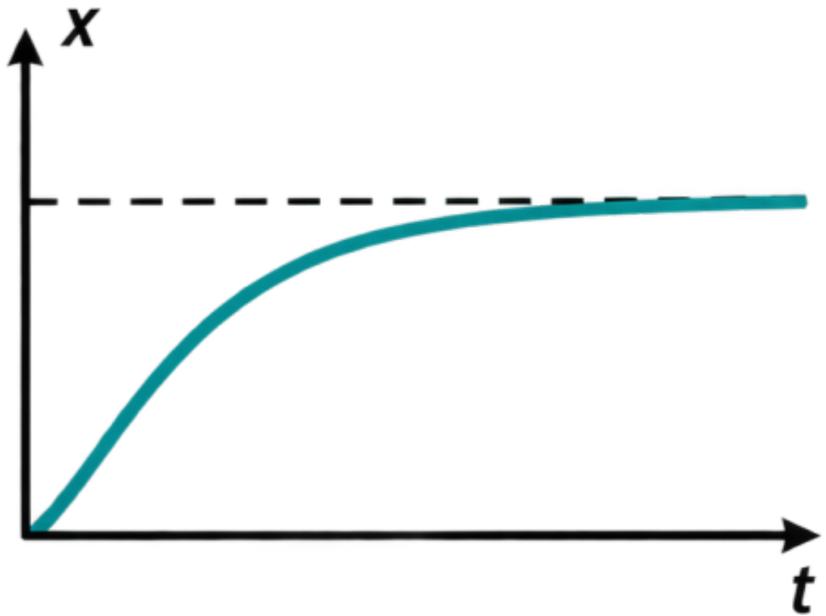

output
A_2
$A_2 - \Delta$
A_1
a_1
a
0
s_1
s_2
stimulus
$-\Delta$
cal. error
real line
calibrated line


Input stimulus
$x(t)$
Δx
t_0
t
Sudden input change
Zero-order response
$y_0(t)$
$K\Delta x$
t_0
t
Instantaneous
First-order response
$y_1(t)$
$K\Delta x$
t_0
t
Gradual response

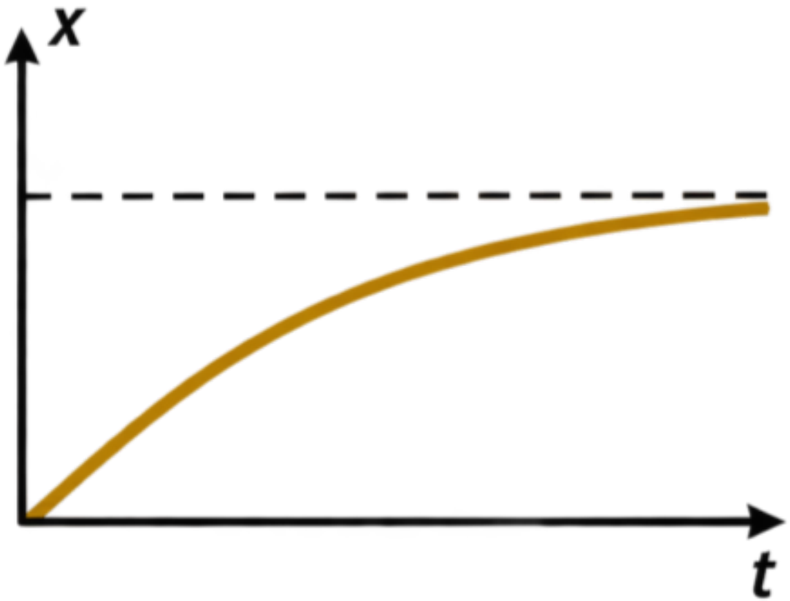

Impulse input
x(t)
δ(t)
∫_{−∞}^{∞} δ(t) dt = 1
0
t
Unit area, zero duration
Impulse response
h(t)
K/τ
h(t) = (K/τ) e^{−t/τ} u(t)
0
t
Step input
x(t)
Δx
x(t) = Δx u(t)
0
t
Sudden increase, constant afterwards
Step response
y(t)
KΔx
y(t) = KΔx(1 − e^{−t/τ}) u(t)
0
t



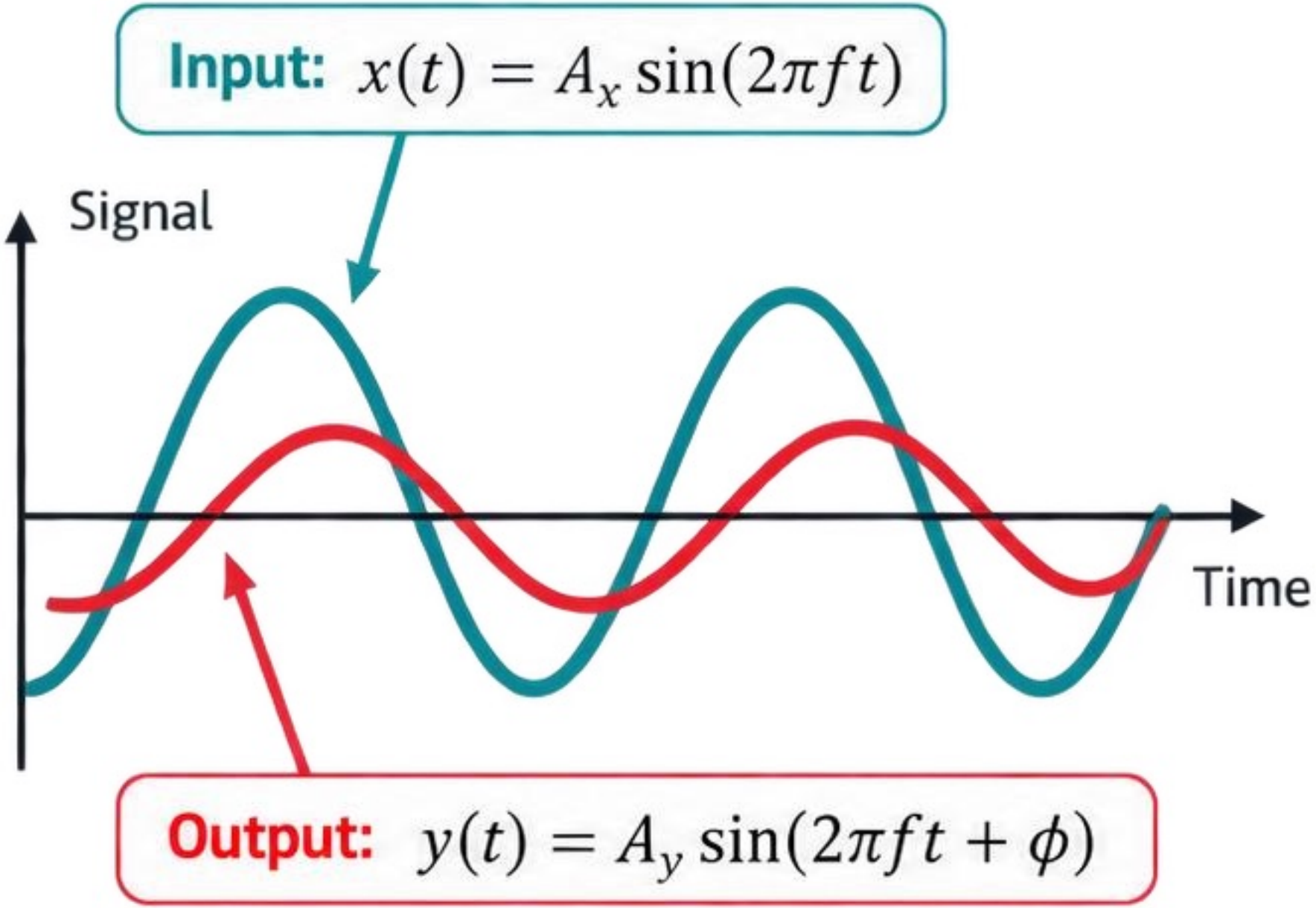
Underdamped

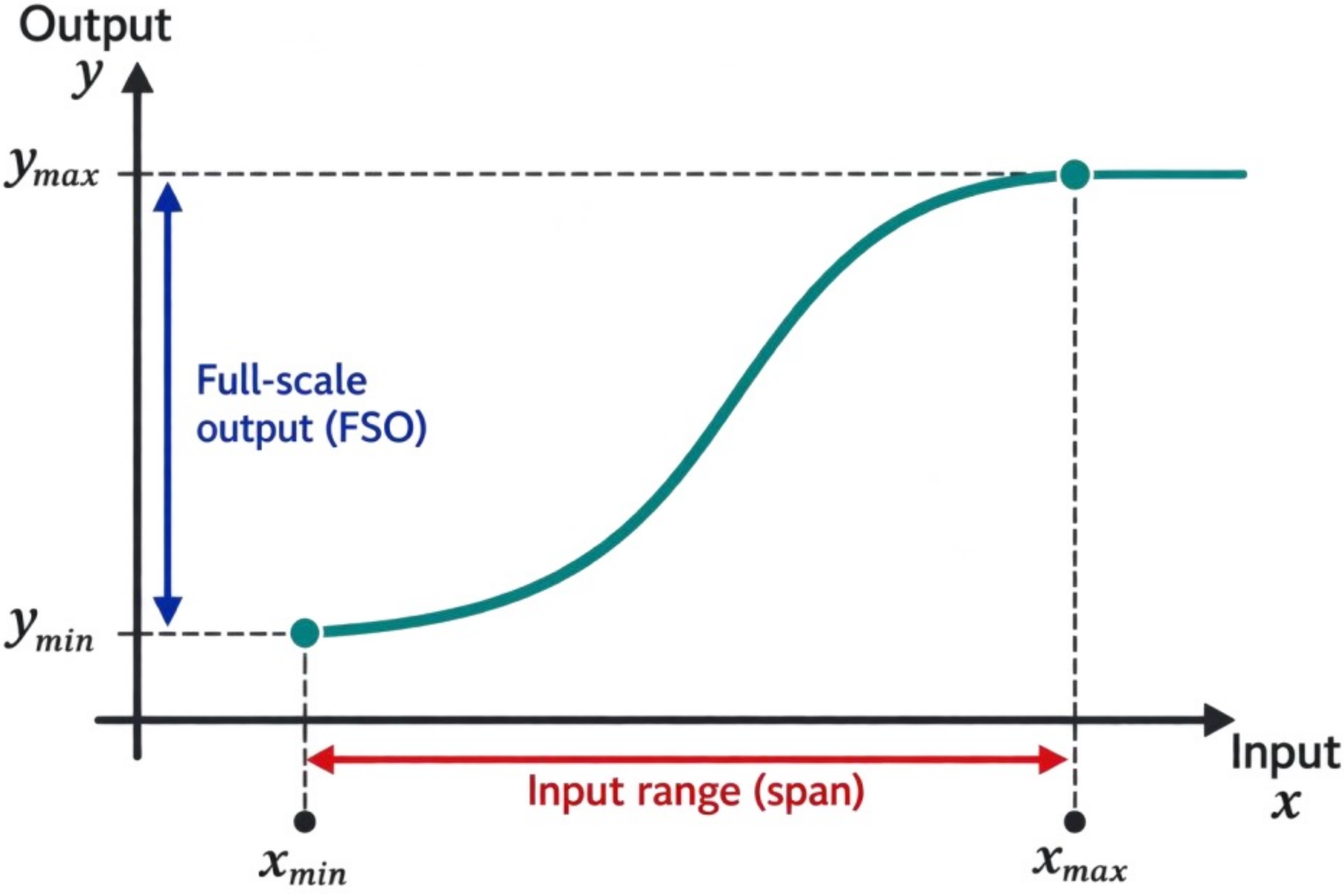


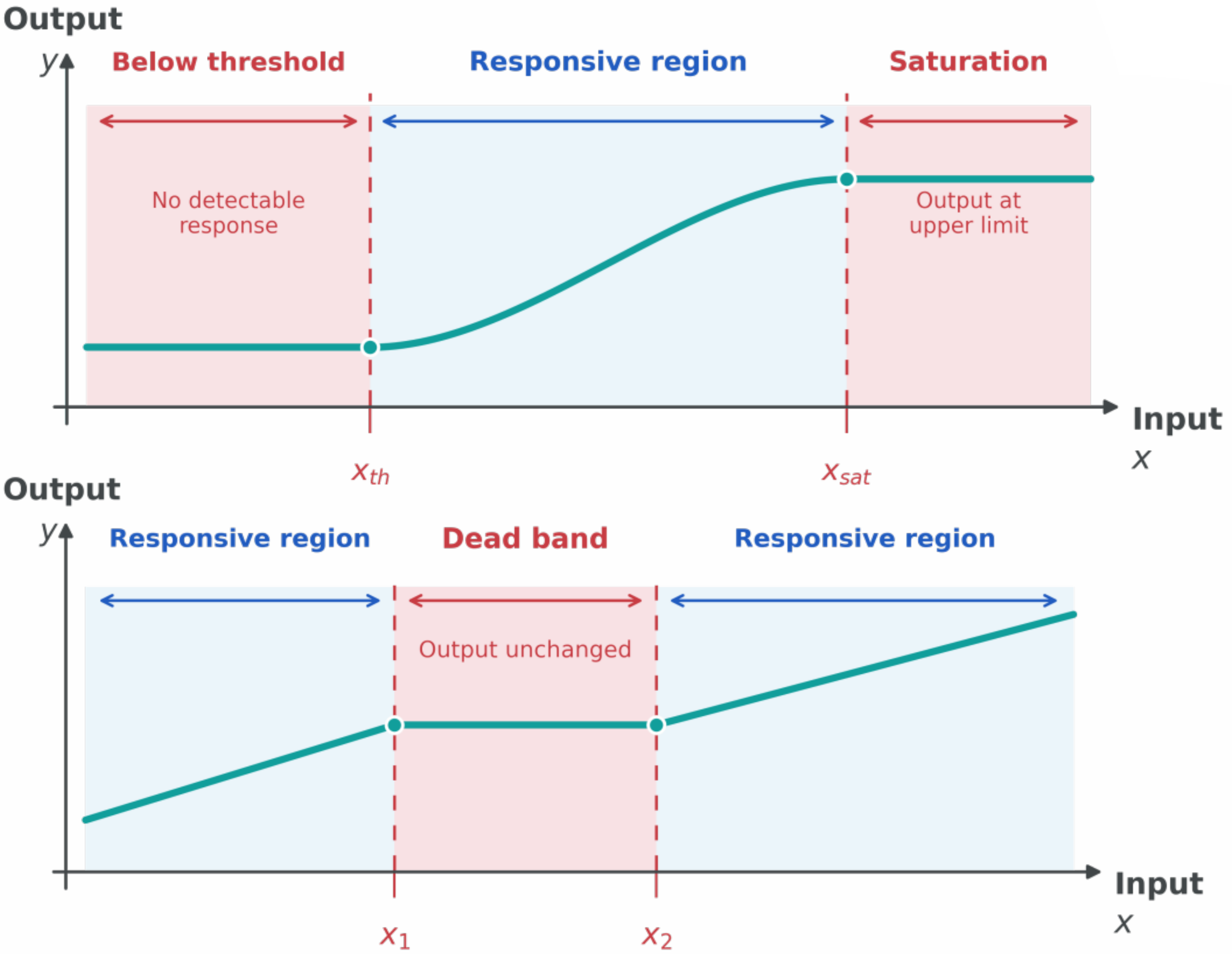
Critically damped

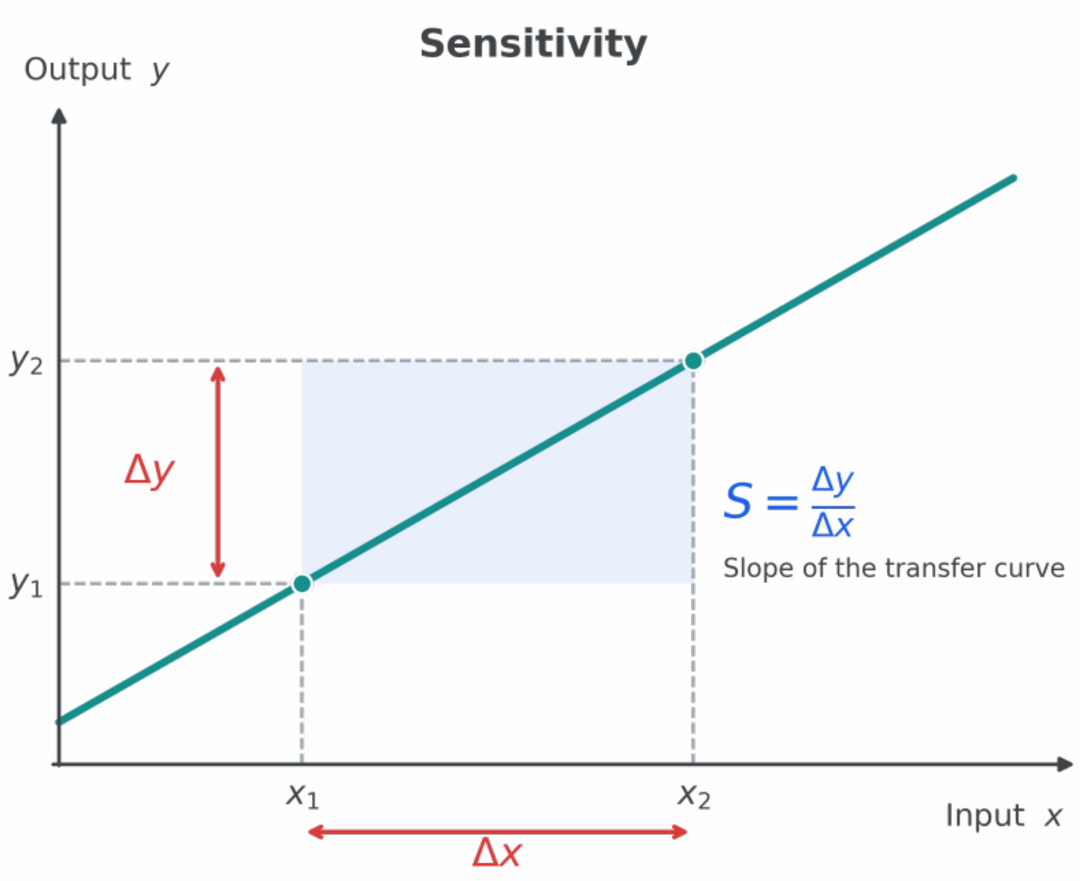


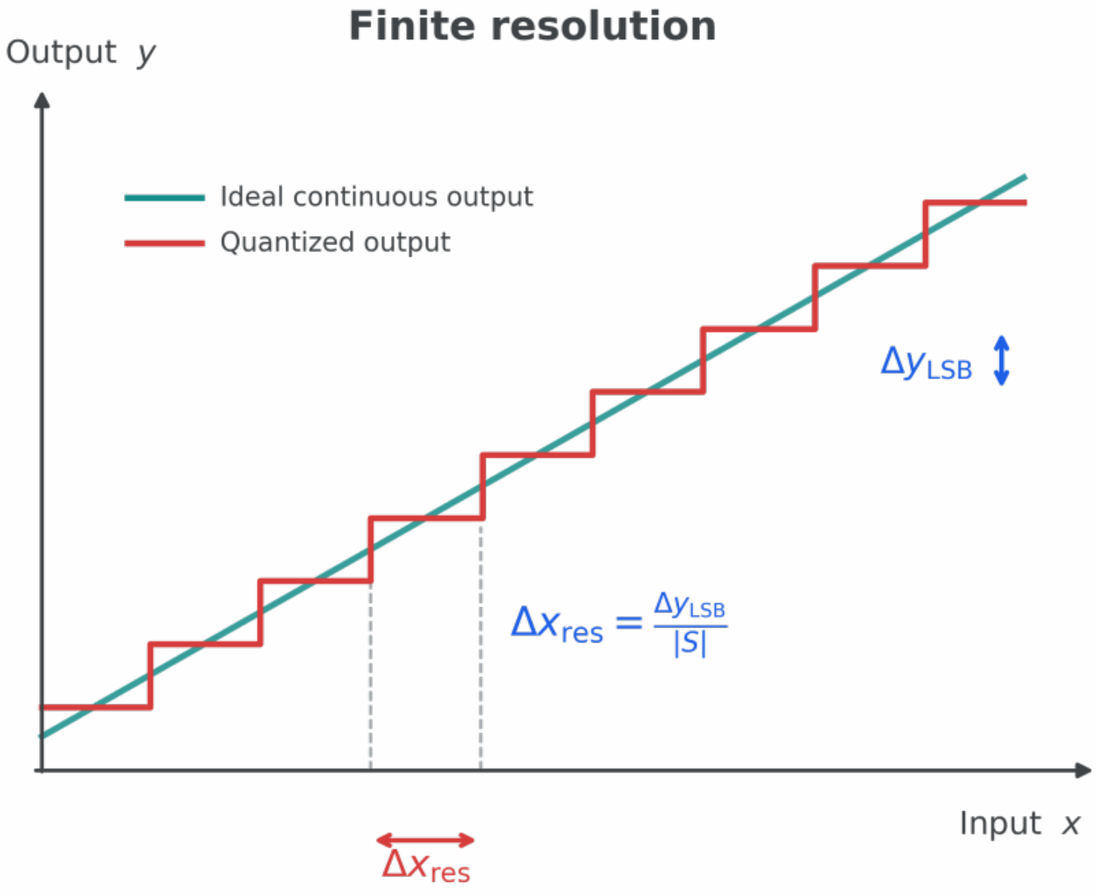
Overdamped

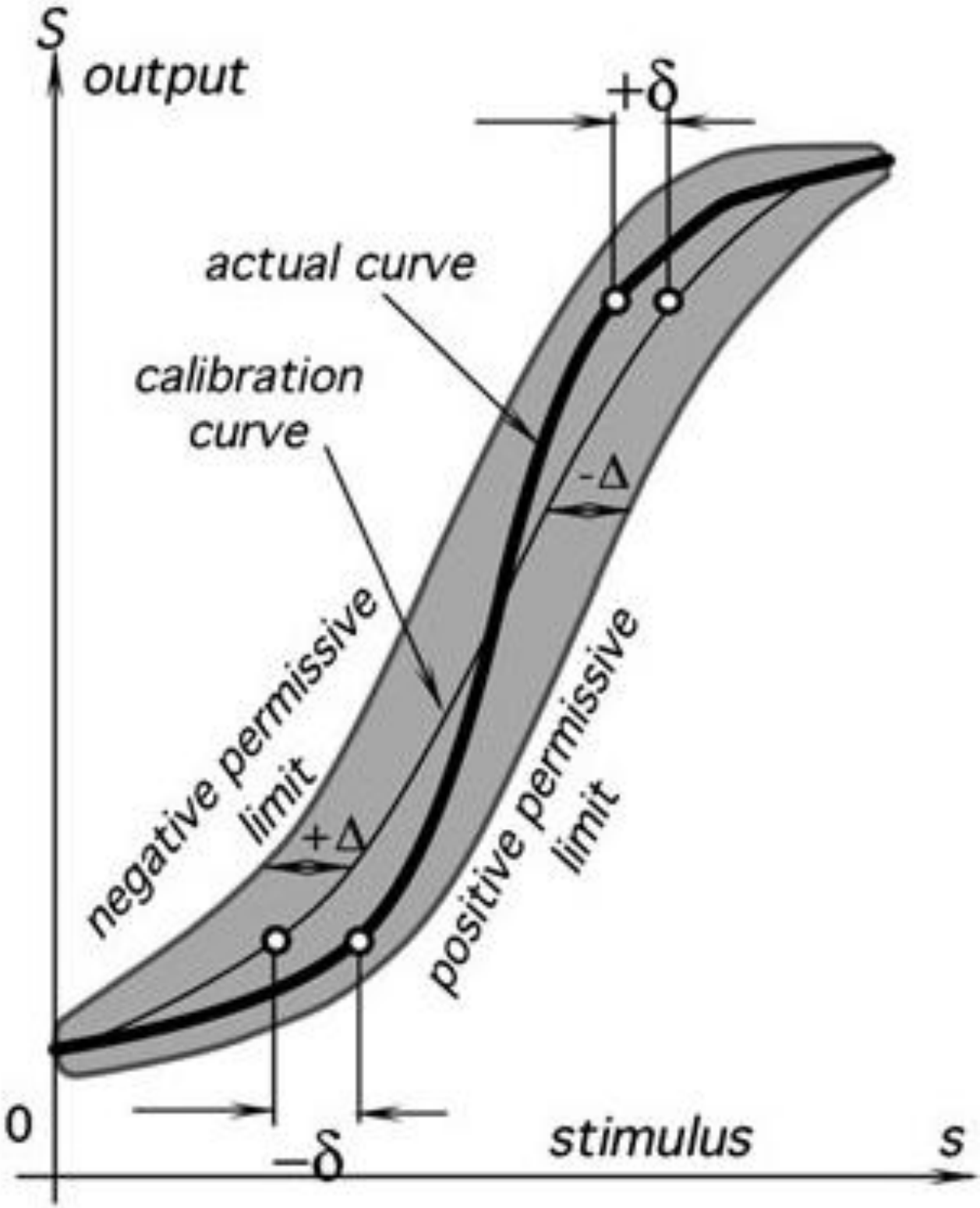

Input: $x(t) = A_x \sin(2\pi f t)$
Signal
Time
Output: $y(t) = A_y \sin(2\pi f t + \phi)$

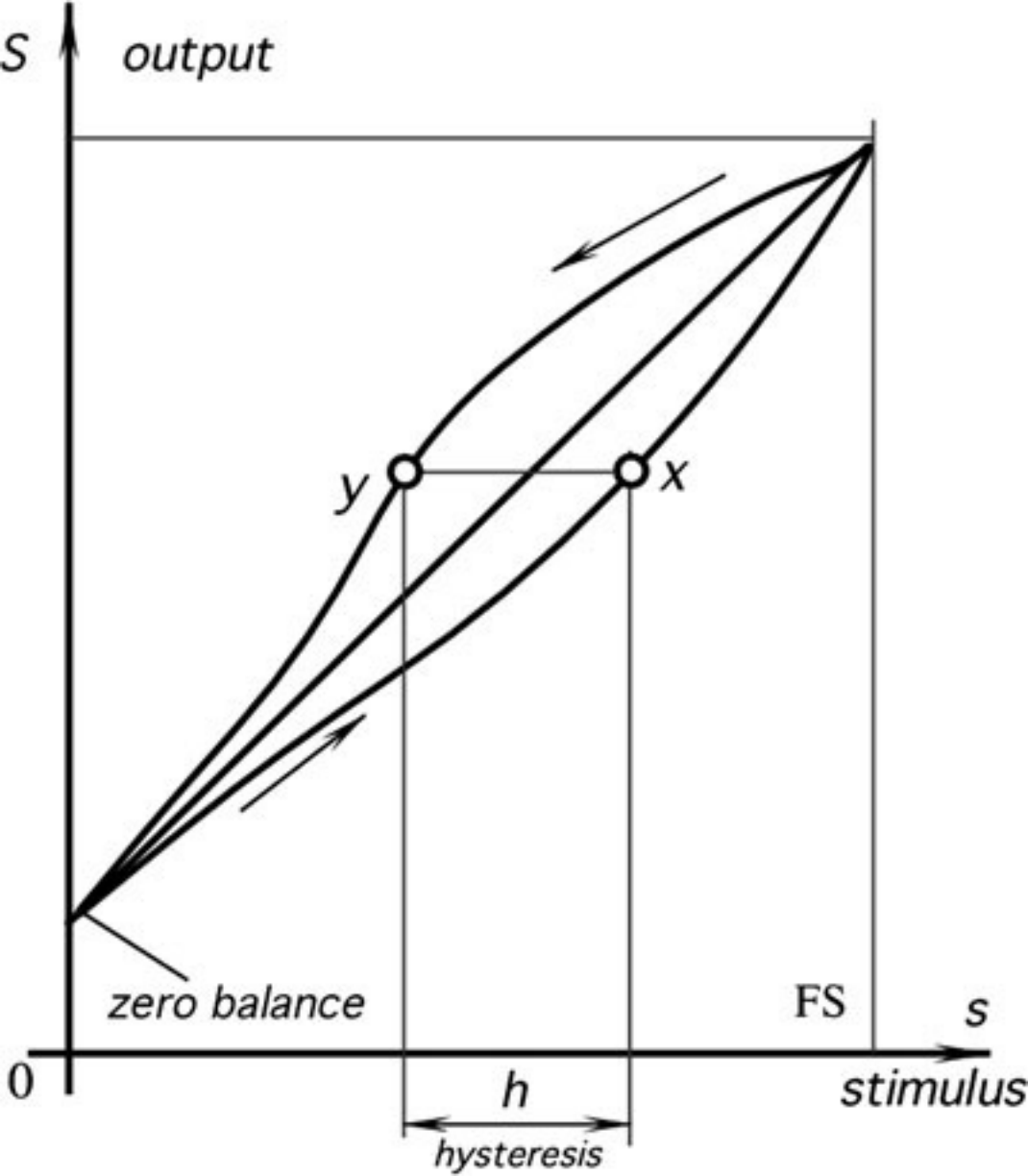

Output
y
y_{max}
Full-scale
output (FSO)
y_{min}
Input range (span)
Input
x
x_{min}
x_{max}


Output
y
Below threshold
Responsive region
Saturation
No detectable response
Output at upper limit
Input
x
x_{th}
x_{sat}
Output
y
Responsive region
Dead band
Responsive region
Output unchanged
Input
x
x_1
x_2

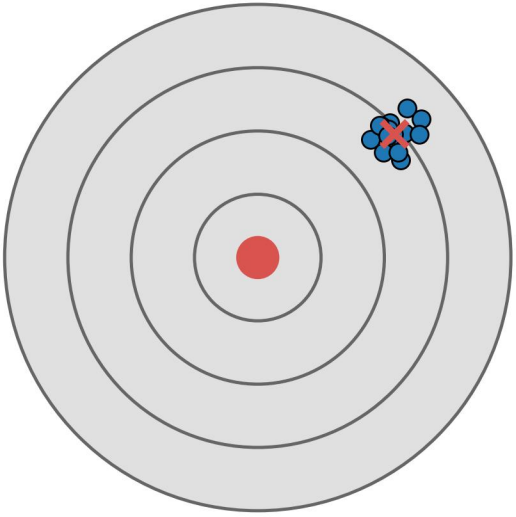

Sensitivity
Output y
y_2
y_1
Δy
$S = \frac{\Delta y}{\Delta x}$
Slope of the transfer curve
x_1
x_2
Δx
Input x


Finite resolution
Output y
Ideal continuous output
Quantized output
Δy_{LSB}
$\Delta x_{res} = \frac{\Delta y_{LSB}}{|S|}$
Δx_{res}
Input x


S
output
+δ
actual curve
calibration
curve
-Δ
negative permissive
limit
+Δ
positive permissive
limit
0
−δ
stimulus
s

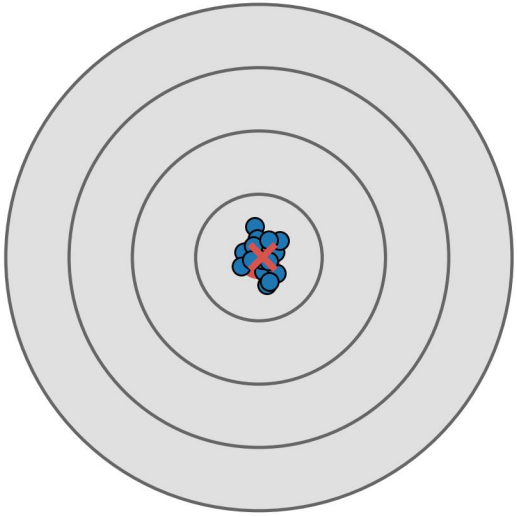

S
output
y
x
zero balance
FS
s
0
h
stimulus
hysteresis

(a) High Precision, Low Trueness



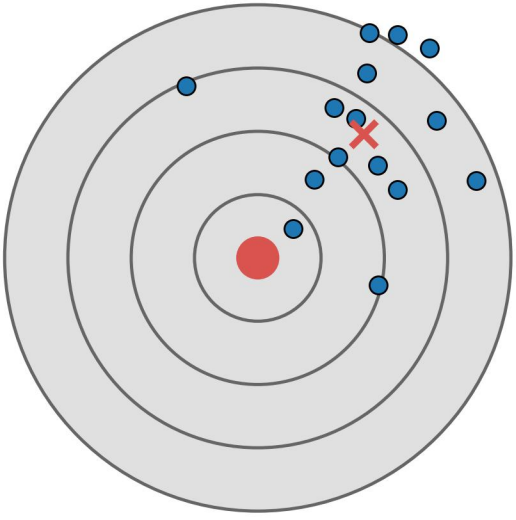
Low Random Error | Large Systematic Bias

(b) High Precision, High Trueness



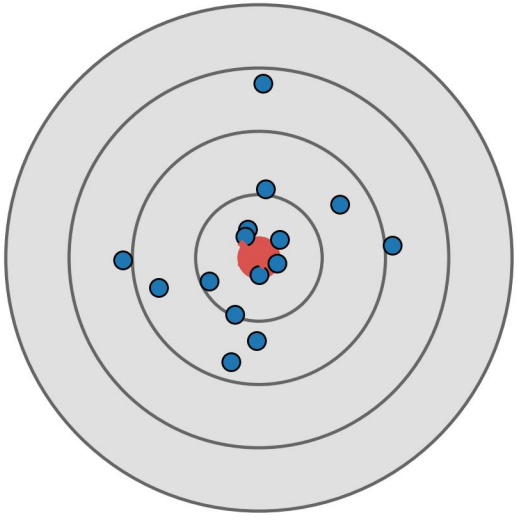
Low Random Error | Low Systematic Bias (High Accuracy)

(c) Low Precision, Low Trueness

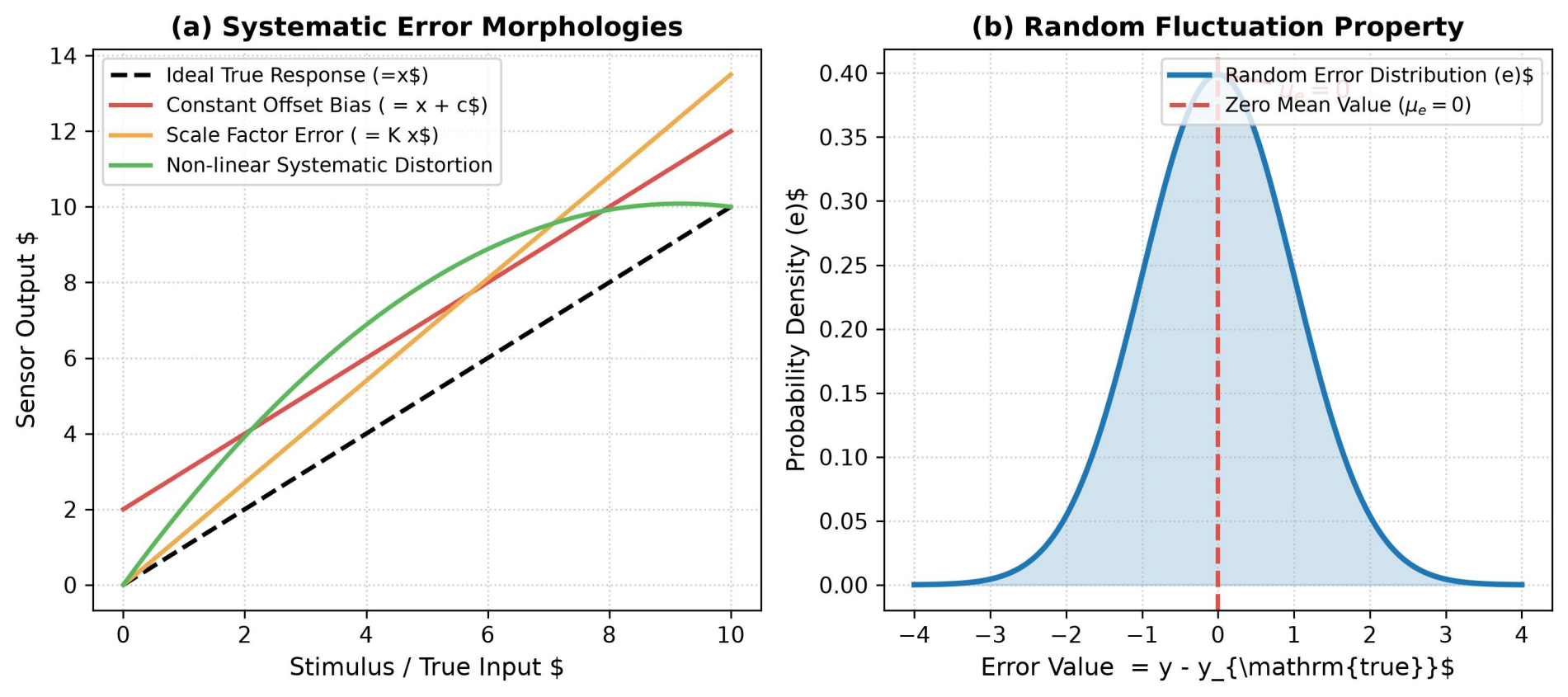


High Random Error | Large Systematic Bias

(d) Low Precision, High Trueness



High Random Error | Low Systematic Bias


(a) Systematic Error Morphologies
Ideal True Response (=x$)
Constant Offset Bias (= x + c$)
Scale Factor Error (= K x$)
Non-linear Systematic Distortion
Sensor Output $
Stimulus / True Input $
0
2
4
6
8
10
12
14
(b) Random Fluctuation Property
Random Error Distribution (e)$
Zero Mean Value ($\mu_e = 0$)
Probability Density (e)$
Error Value = y - y_{\mathrm{true}}$
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
−4
−3
−2
−1
0
1
2
3
4